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FLAT TOP GAS GRILLS INCLUDING COOKING ENGINES CONFIGURED FOR OPTIMUM HEAT DISTRIBUTION

Abstract

Flat top gas grills including waste collection assemblies are disclosed. A cookbox of a grill includes a bottom wall and a lower waste disposal opening extending through the bottom wall. A griddle is disposed on or above the cookbox and includes an upper waste disposal opening extending through a base of the griddle. Burner tubes are disposed in the cookbox between the bottom wall of the cookbox and the base of the griddle. A waste bin is slidably coupled to the cookbox and located below the lower waste disposal opening of the cookbox. A waste disposal chute is disposed within the cookbox. The waste disposal chute is configured to receive waste from the upper waste disposal opening of the griddle and direct the received waste to pass through the lower waste disposal opening of the cookbox and into the waste bin.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/380,722, filed Oct. 17, 2023. The entirety of U.S. patent application Ser. No. 18/380,722 is hereby incorporated by reference herein.

[0002] FIELD OF THE DISCLOSURE

[0003] This disclosure relates generally to flat top gas grills and, more specifically, to flat top gas grills including cooking engines configured for optimum heat distribution.

BACKGROUND

[0004] Many conventional gas grills are equipped with grated cooking surfaces formed by a plurality of interconnected rungs and/or grate members that are spaced apart from one another to form a framework having many openings. Unlike gas grills that are equipped with such grated cooking surfaces, flat top gas grills instead include a griddle configured as a generally continuous and flat cooking surface that optionally includes a small number of openings (e.g., typically one or two openings) extending therethrough, with such openings being configured to receive grease, residue, particles, and/or other byproducts associated with a cooking operation performed on the griddle. While a griddle can be used to cook many different types and/or sizes of food items, implementing a griddle as the cooking surface of the gas grill can be particularly advantageous when it comes to cooking liquid or semi-liquid food items (e.g., eggs, batter, etc.) and/or small pieces of food items (e.g., ground meat, diced or chopped vegetables, etc.) that would otherwise be prone to falling through one or more of the many openings of a typical grated cooking surface.

[0005] The cooking engine of a flat top gas grill conventionally includes a cookbox, a griddle disposed on or above the cookbox, and one or more gas-fueled burner tube(s) disposed in the cookbox at a location between a bottom wall of the cookbox and an underside of the griddle. Cooking operations performed on flat top gas grills are typically carried out with a lid of the grill either removed from the cookbox or placed in an open position relative to the cookbox. The absence of a lid during cooking operations can lead to substantial heat loss from the cookbox and/or the griddle, and can also lead to the formation of significant heat distribution and/or temperature variations (e.g., the presence of cold spots and/or hot spots) across the cooking surface of the griddle. In many commercially available flat top gas grills, such variations can exceed two hundred degrees Fahrenheit (200° F.), even in instances when all of the burner tubes are adjusted to the same output setting. Consumers often find such extreme variations to be undesirable and/or unacceptable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a perspective view of an example grill constructed in accordance with the teachings of this disclosure.

[0007] FIG. 2 is an exploded view of the grill of FIG. 1.

[0008] FIG. 3 is a perspective view showing the cookbox of the grill of FIGS. 1 and 2 in isolation.

[0009] FIG. 4 is a top view of the cookbox of FIG. 3.

[0010] FIG. 5 is a first perspective view showing the griddle of the grill of FIGS. 1 and 2 in isolation.

[0011] FIG. **6** is a second perspective view of the griddle of FIG. **5**.

[0012] FIG. **7** is a perspective view showing the first burner tube of the grill of FIGS. **1** and **2** in isolation.

[0013] FIG. **8** is a top view of the first burner tube of FIG. **7**.

[0014] FIG. **9** is a bottom view of the first burner tube of FIGS. **7** and **8**.

[0015] FIG. **10** is a right side view of the first burner tube of FIGS. **7-9**.

[0016] FIG. **11** is a cross-sectional view of the first burner tube of FIGS. **7-10** taken along section A-A of FIG. **8**.

[0017] FIG. **12** is a front view showing the grill of FIGS. **1** and **2**, with the frame, the lid, and certain other structural features of the grill omitted for enhanced viewability.

[0018] FIG. **13** is a right side view of the grill of FIG. **12**.

[0019] FIG. **14** is a top view of the grill of FIGS. **12** and **13**.

[0020] FIG. **15** is a cross-sectional view of the grill of FIGS. **12-14** taken along section B-B of FIG. **14**.

[0021] FIG. **16** is a cross-sectional view of the grill of FIGS. **12-15** taken along section C-C of FIG. **14**.

[0022] FIG. **17** is a cross-sectional view of the grill of FIGS. **12-16** taken along section D-D of FIG. **12**.

[0023] FIG. **18** is a perspective view of another example grill constructed in accordance with the teachings of this disclosure.

[0024] FIG. **19** is an exploded view of the grill of FIG. **18**.

[0025] FIG. **20** is a perspective view showing the cookbox of the grill of FIGS. **18** and **19** in isolation.

[0026] FIG. **21** is a top view of the cookbox of FIG. **20**.

[0027] FIG. **22** is a first perspective view showing the griddle of the grill of FIGS. **18** and **19** in isolation.

[0028] FIG. **23** is a second perspective view of the griddle of FIG. **22**.

[0029] FIG. **24** is a perspective view showing the first burner tube of the grill of FIGS. **18** and **19** in isolation.

[0030] FIG. **25** is a top view of the first burner tube of FIG. **24**.

[0031] FIG. **26** is a bottom view of the first burner tube of FIGS. **24** and **25**.

[0032] FIG. **27** is a right side view of the first burner tube of FIGS. **24-26**.

[0033] FIG. **28** is a cross-sectional view of the first burner tube of FIGS. **24-27** taken along section E-E of FIG. **25**.

[0034] FIG. **29** is a front view showing the grill of FIGS. **18** and **19**, with the frame, the lid, and certain other structural features of the grill omitted for enhanced viewability.

[0035] FIG. **30** is a right side view of the grill of FIG. **29**.

[0036] FIG. **31** is a top view of the grill of FIGS. **29** and **30**.

[0037] FIG. **32** is a cross-sectional view of the grill of FIGS. **29-31** taken along section F-F of FIG. **31**.

[0038] FIG. **33** is a cross-sectional view of the grill of FIGS. **29-32** taken along section G-G of FIG. **31**.

[0039] FIG. **34** is a cross-sectional view of the grill of FIGS. **29-33** taken along section H-H of FIG. **29**.

[0040] Certain examples are shown in the above-identified figures and described in detail below. In describing these examples, like or identical reference numbers are used to identify the same or similar elements. The figures are not necessarily to scale and certain features and certain views of the figures may be shown exaggerated in scale or in schematic for clarity and/or conciseness.

[0041] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order,

arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name.

DETAILED DESCRIPTION

[0042] As discussed above, the cooking engine of a flat top gas grill conventionally includes a cookbox, a griddle disposed on or above the cookbox, and one or more gas-fueled burner tube(s) disposed in the cookbox at a location between a bottom wall of the cookbox and an underside of the griddle. Cooking operations performed on flat top gas grills are typically carried out with a lid of the grill either removed from the cookbox or placed in an open position relative to the cookbox. The absence of a lid during cooking operations can lead to substantial heat loss from the cookbox and/or the griddle, and can also lead to the formation of significant heat distribution and/or temperature variations (e.g., the presence of cold spots and/or hot spots) across the cooking surface of the griddle.

[0043] For example, cooking operations performed on many commercially-available flat top gas grills can generate heat distribution and/or temperature variations in excess of two hundred degrees Fahrenheit (200° F.) across the cooking surface of the griddle, even in instances when all of the gas-fueled burner tubes of the flat top gas grill are adjusted to the same output setting (e.g., all burner tubes adjusted to a LOW output setting, all burner tubes adjusted to a MEDIUM output setting, or all burner tubes adjusted to a HI output setting). In addition to making the cooking process highly inefficient, heat distribution and/or temperature variations of such a stark magnitude can also effectively render portions of the cooking surface of the griddle unusable for their intended purpose, thereby resulting in a less than optimum user experience for the consumer.

[0044] Example flat top gas grills disclosed herein include cooking engines configured for optimum heat distribution. In some disclosed examples, a grill includes: (1) a cookbox having a plurality of air intake openings formed in and extending through a bottom wall of the cookbox; (2) a griddle disposed on or above the cookbox, the griddle including a flat top cooking surface and an underside located opposite the flat top cooking surface; and (3) a plurality of burner tubes disposed in the cookbox and located between the bottom wall of the cookbox and the underside of the griddle. In some disclosed examples, the cookbox of the grill is advantageously configured such that an unoccupied area collectively defined by the plurality of air intake openings is between 10.0 and 30.0 percent of the total area of the bottom wall of the cookbox. In some disclosed examples, the cookbox and the burner tubes of the grill are advantageously configured such that respective ones of the air intake openings are laterally positioned between neighboring ones of the burner tubes, and/or such that respective ones of the burner tubes are laterally aligned with a solid portion of the bottom wall of the cookbox that is laterally positioned between neighboring ones of the air intake openings. In some disclosed examples, the cookbox and the griddle of the grill are advantageously configured such that a vertical gap extending between an upper rim of the cookbox and the underside of the griddle is between 0.25 and 1.50 inches, wherein the vertical gap is configured to vent heat generated within the cookbox (e.g., heat generated by one or more of the burner tube(s)).

[0045] In some disclosed examples, the cookbox, the griddle, and one or more of the burner tube(s) of the grill are advantageously configured such that a central axis of the burner tube is located between 1.5 and 3.5 inches above the bottom wall of the cookbox, located between 2.0 and 4.0 inches below an upper rim of the cookbox, and/or located between 3.0 and 5.0 inches below the underside of the griddle. In some disclosed examples, one or more of the burner tube(s) of the grill include(s) first and second rows of ports extending through an outer wall of the burner tube and respectively arranged parallel to the central axis of the burner tube, wherein the first and second

rows of ports are advantageously configured to be angularly displaced from one another by an angle between 90.0 and 150.0 degrees. In some disclosed examples, one or more of the burner tube(s) of the grill include(s) third and fourth rows of ports extending through the outer wall of the burner tube and respectively arranged perpendicular to the central axis of the burner tube, wherein the third row of ports is advantageously configured to be located between 1.0 and 6.0 inches inwardly from a front wall of the cookbox, and wherein the fourth row of ports is advantageously configured to be located between 1.0 and 6.0 inches inwardly from a rear wall of the cookbox.

[0046] The above-described features implemented by the disclosed flat top gas grills advantageously improve the operating efficiency of the cooking engine of the flat top gas grill, and advantageously improve the heat distribution properties associated with the griddle of the flat top gas grill. In this regard, the above-described features individually and collectively assist in minimizing any temperature variation across the flat top cooking surface of the griddle during cooking operations performed thereon. Flat top gas grills disclosed herein maintain temperature variations of approximately 150 degrees Fahrenheit (150° F.) or less across the flat top cooking surface of the griddle when the burner tubes of the grill are adjusted to a common output setting. Such temperature variations are substantially less than those found in many commercially available flat top gas grills, some of which produce flat top cooking surface temperature variations exceeding two hundred degrees Fahrenheit (200° F.) when the burner tubes of the flat top gas grill are adjusted to a common output setting.

[0047] The above-identified features as well as other advantageous features of example flat top gas grills including cooking engines configured for optimum heat distribution as disclosed herein are further described below in connection with the figures of the application.

[0048] As used herein, the term “configured” means sized, shaped, arranged, structured, oriented, positioned, and/or located. For example, in the context of a first part configured to fit within a second part, the first part is sized, shaped, arranged, structured, oriented, positioned, and/or located to fit within the second part.

[0049] As used herein, unless otherwise stated, the terms “above” and “below” describe the relationship of two parts relative to Earth. For example, as used herein, a first part is “above” a second part if the second part is closer to Earth than the first part is. As another example, as used herein, a first part is “below” a second part if the first part is closer to Earth than the second part is. It is to be understood that a first part can be above or below a second part with one or more of: another part or parts therebetween; without another part therebetween; with the first and second parts contacting one another; or without the first and second parts contacting one another.

[0050] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0051] As used herein, the terms “substantially” and/or “approximately” modify their subjects and/or values to recognize the potential presence of variations that occur in real world applications. For example, “substantially” and/or “approximately” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real-world imperfections as will be understood by persons of ordinary skill in the art. For example, “substantially” and/or “approximately” may indicate such dimensions may be within a tolerance range of +/-10% unless otherwise specified in the description provided herein.

[0052] As used herein, the terms “including” and “comprising” (and all forms and tenses thereof) are open-ended terms. Thus, whenever the written description or a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements,

terms, etc., may be present without falling outside the scope of the corresponding claim or recitation.

[0053] As used herein, singular references (e.g., “a,” “an,” “first,” “second,” etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein.

Furthermore, although individually listed, a plurality of means, elements, or method actions may be implemented by, for example, the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0054] The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C.

[0055] As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open-ended. As used herein in the context of describing structures, components, items, objects, and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects, and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0056] FIG. 1 is a perspective view of an example grill **100** constructed in accordance with the teachings of this disclosure. FIG. 2 is an exploded view of the grill **100** of FIG. 1. The grill **100** of FIGS. 1 and 2 is structured as a flat top gas grill. In this regard, the grill **100** of FIGS. 1 and 2 includes an example cookbox **102** and an example griddle **104**. The griddle **104** of the grill **100** is configured to be disposed on or above the cookbox **102** of the grill **100**. The cookbox **102** of the grill **100** is further described below in connection with FIGS. 3 and 4. The griddle **104** of the grill **100** is further described below in connection with FIGS. 5 and 6.

[0057] The grill **100** of FIGS. 1 and 2 further includes a plurality of burner tubes configured to be disposed within the cookbox **102** at a location between a bottom wall of the cookbox **102** and an upper rim defined by or associated with the cookbox **102**, and/or at a location between the bottom wall of the cookbox **102** and an underside of the griddle **104**. As shown in FIG. 2, the plurality of burner tubes includes an example first burner tube **202**, an example second burner tube **204**, and an example third burner tube **206** (e.g., a total of three burner tubes) that are configured to be laterally spaced apart from one another and arranged in a front-to-rear orientation within the cookbox **102**, with the first burner tube **202**, the second burner tube **204**, and the third burner tube **206** being of a substantially identical construction relative to one another. In other examples, the plurality of burner tubes can instead include a different number (e.g., two, four, five, etc.) of burner tubes, and the construction of one or more of the burner tubes may differ from that of the first burner tube **202**, the second burner tube **204**, and/or the third burner tube **206** shown in FIG. 2. The first burner tube **202** of the grill **100** is further described below in connection with FIGS. 7-11.

[0058] In the illustrated example of FIGS. 1 and 2, the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) form part of a gas train that further includes an example fuel source **106**, an example regulator assembly **208**, an example manifold

210, and a plurality of control valves corresponding in number to the plurality of burner tubes. In this regard, the plurality of control valves of the grill **100** as shown in FIG. 2 includes an example first control valve **212** associated with the first burner tube **202**, an example second control valve **214** associated with the second burner tube **204**, and an example third control valve **216** associated with the third burner tube **206**.

[0059] In the illustrated example of FIG. 1, the fuel source **106** is implemented as a fuel tank (e.g., a propane tank) containing combustible gas. In other examples, the fuel source **106** can instead be implemented as a piped (e.g., household) natural gas line that provides an accessible flow of combustible gas. The regulator assembly **208** is operatively positioned between the fuel source **106** and the manifold **210** such that a supply of combustible gas provided via the fuel source **106** flows through the regulator assembly **208** and into the manifold **210**. The first control valve **212** is operatively positioned between the manifold **210** and the first burner tube **202** such that combustible gas received at the manifold **210** can be selectively supplied to the first burner tube **202** via the first control valve **212**. The second control valve **214** is operatively positioned between the manifold **210** and the second burner tube **204** such that combustible gas received at the manifold **210** can be selectively supplied to the second burner tube **204** via the second control valve **214**. The third control valve **216** is operatively positioned between the manifold **210** and the third burner tube **206** such that combustible gas received at the manifold **210** can be selectively supplied to the third burner tube **206** via the third control valve **216**.

[0060] In the illustrated example of FIGS. 1 and 2, the manifold **210**, the first control valve **212**, the second control valve **214**, and the third control valve **216** are at least partially covered and/or concealed by an example control panel **108** that is coupled to and/or located along the front of the cookbox **102**. The grill **100** further includes a plurality of control knobs mounted and/or located along the front face of the control panel **108**, with the plurality of control knobs corresponding in number to the plurality of control valves and/or the number of burner tubes. In this regard, the plurality of control valves of the grill **100** as shown in FIGS. 1 and 2 includes an example first control knob **110** associated with the first control valve **212** and/or the first burner tube **202**, an example second control knob **112** associated with the second control valve **214** and/or the second burner tube **204**, and an example third control knob **114** associated with the third control valve **216** and/or the third burner tube **206**. Each control knob is mechanically coupled to its corresponding control valve such that movement (e.g., rotation) of the control knob changes the extent to which an adjustable flow control member of the corresponding control valve enables combustible gas to flow through the corresponding control valve into the corresponding burner tube. For example, the first control knob **110** is mechanically coupled to the first control valve **212** such that movement (e.g., rotation) of the first control knob **110** changes the extent to which an adjustable flow control member of the first control valve **212** enables combustible gas to flow through the first control valve **212** into the first burner tube **202**.

[0061] The grill **100** of FIGS. 1 and 2 further includes an example waste collection assembly **116** configured to collect and facilitate the removal of cooking waste. In the illustrated example of FIGS. 1 and 2, the waste collection assembly **116** includes an example waste bin **118** that is suspended from and slidably coupled to a pair of example support rails **120**, with the support rails **120** being coupled to an underside of the cookbox **102**. The waste bin **118** is moveable (e.g., slidable) along the support rails **120** between a closed position in which a substantial portion of the waste bin **118** is covered by the underside of the cookbox **102** (e.g., as shown in FIG. 1), and an open position in which a substantial portion of the waste bin **118** is located forward of and not covered by the underside of the cookbox **102**. The waste bin **118** is configured to hold and/or contain an example disposable liner **218**.

[0062] When the waste bin **118** is in the closed position, the waste bin **118** and/or the disposable liner **218** is/are positioned below a lower waste disposal opening formed in and extending through a bottom wall of the cookbox **102**. The waste collection assembly **116** further includes an example

waste disposal chute **220** disposed within the cookbox **102**, and an example upper waste disposal opening **122** formed in and extending through the griddle **104**. The waste disposal chute **220** is operatively positioned between the upper waste disposal opening **122** and the lower waste disposal opening such that liquid and/or solid cooking waste (e.g., grease, residue, particles, and/or other byproducts associated with a cooking operation) located on a flat top cooking surface of the griddle **104** can be fed into the upper waste disposal opening **122**, with the cooking waste thereafter passing from the upper waste disposal opening **122** through the waste disposal chute **220** and through the lower waste disposal opening. The waste bin **118** and/or the disposable liner **218** collect(s) cooking waste that passes through the waste disposal chute **220** and the lower waste disposal opening when the waste bin **118** is in the closed position. Moving the waste bin **118** from the closed position into the open position facilitates removal of the disposable liner **218** and/or the cooking waste contained therein.

[0063] The grill **100** of FIGS. **1** and **2** further includes an example frame **124**. In the illustrated example of FIGS. **1** and **2**, the frame **124** includes one or more example cookbox support member(s) **126** (e.g., one or more vertically oriented leg(s)) that are configured to support the cookbox **102** above an underlying ground surface. The cookbox support member(s) **126** and/or, more generally, the frame **124** can be configured from any number and any type of structural components arranged in any manner that facilitates supporting the cookbox **102** above an underlying ground surface when the grill **100** is in use. For example, while each of the one or more cookbox support member(s) **126** shown in FIGS. **1** and **2** is fixed relative to the cookbox **102**, in other examples the frame **124** can include one or more foldable, slidable, and/or telescoping cookbox support member(s) **126** that facilitate collapsing and/or otherwise modifying the frame **124** of the grill **100** when the grill **100** is not in use. The grill **100** of FIGS. **1** and **2** can further include one or more side table(s) coupled to the frame **124** and/or to the cookbox **102** of the grill **100**.

[0064] The grill **100** of FIGS. **1** and **2** further includes an example lid **128** configured to cover and/or enclose the griddle **104** of the grill **100** when the lid **128** is in a closed position. The lid **128** is movable relative to the griddle **104** and/or the cookbox **102** between a closed position and an open position in which the flat top cooking surface of the griddle **104** is exposed (e.g., as shown in FIG. **1**). In the illustrated example of FIGS. **1** and **2**, the lid **128** is removably positioned on the griddle **104** and/or the cookbox **102** without there being any direct mechanical coupling between the lid **128** on the one hand and the cookbox **102**, the griddle **104**, and/or the frame **124** on the other hand. In other examples, the lid **128** can be pivoted relative to the cookbox **102**, the griddle **104**, and/or the frame **124** via one or more hinge(s) that mechanically couple the lid **128** to the cookbox **102**, the griddle **104**, and/or the frame **124**. Movement of the lid **128** of the grill **100** between the closed position and the open position can be facilitated via user interaction with an example handle **130** that is coupled to the lid **128**.

[0065] FIG. **3** is a perspective view showing the cookbox **102** of the grill **100** of FIGS. **1** and **2** in isolation. FIG. **4** is a top view of the cookbox **102** of FIG. **3**. As shown in FIGS. **3** and **4**, the cookbox **102** includes an example bottom wall **302**, an example front wall **304**, an example rear wall **306**, an example right sidewall **308**, and an example left sidewall **310**. The rear wall **306** of the cookbox **102** is located opposite the front wall **304** of the cookbox **102**. The right sidewall **308** of the cookbox **102** extends between the front wall **304** and the rear wall **306** of the cookbox **102**. The left sidewall **310** of the cookbox **102** is located opposite the right sidewall **308** of the cookbox **102** and extends between the front wall **304** and the rear wall **306** of the cookbox **102**. As shown in FIGS. **3** and **4**, the front wall **304**, the rear wall **306**, the right sidewall **308**, and the left sidewall **310** of the cookbox **102** extend upwardly from the bottom wall **302** of the cookbox **102** to define a cavity in which one or more structural component(s) (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) of the grill **100** can be disposed.

[0066] The cookbox **102** further includes a plurality of air intake openings formed in and extending

through the bottom wall **302** of the cookbox **102**. The air intake openings are individually and collectively configured to draw air from the surrounding atmosphere into the cookbox **102**. As shown in FIGS. **3** and **4**, the plurality of air intake openings includes an example first air intake opening **312** and an example second air intake opening **314**, with the second air intake opening **314** being laterally spaced apart from the first air intake opening **312**. In this regard, the bottom wall **302** of the cookbox **102** further includes an example solid portion **316** that extends between the front wall **304** and the rear wall **306** of the cookbox **102**, and further extends between the first air intake opening **312** and the second air intake opening **314** formed in the bottom wall **302** of the cookbox **102**. In the illustrated example of FIGS. **3** and **4**, the solid portion **316** is free of any unplugged and/or uncovered opening(s) that extend through the bottom wall **302**.

[0067] The first air intake opening **312**, the second air intake opening **314**, and the solid portion **316** of the bottom wall **302** of the cookbox **102** are individually and collectively configured such that a burner tube (e.g., the second burner tube **204**) of the grill **100** is laterally aligned with the solid portion **316**, and such that said burner tube (e.g., the second burner tube **204**) is laterally positioned between the first air intake opening **312** and the second air intake opening **314**. The first air intake opening **312**, the second air intake opening **314**, and the solid portion **316** of the bottom wall **302** of the cookbox **102** are further individually and collectively configured such that the first air intake opening **312** is laterally positioned between one set of neighboring burner tubes (e.g., the first burner tube **202** and the second burner tube **204**) of the grill **100**, and such that the second air intake opening **314** is laterally positioned between another set of neighboring burner tubes (e.g., the second burner tube **204** and the third burner tube **206**) of the grill **100**.

[0068] In the illustrated example of FIGS. **3** and **4**, the plurality of air intake openings includes a total of two air intake openings (e.g., the first air intake opening **312** and the second air intake opening **314**). In other examples, the plurality of air intake openings can instead include a different number (e.g., three, four, five, etc.) of air intake openings. In the illustrated example of FIGS. **3** and **4**, the first air intake opening **312** and the second air intake opening **314** each have a generally rectangular shape. In other examples, the first air intake opening **312** and/or the second air intake opening **314** can instead have a different shape (e.g., a circular shape, an oval shape, a triangular shape, a trapezoidal shape, etc.).

[0069] In addition to the air intake openings illustrated in FIGS. **3** and **4**, the cookbox **102** further includes an example lower waste disposal opening **402** formed in and extending through the bottom wall **302** of the cookbox **102**. The lower waste disposal opening **402** is configured to receive and/or to otherwise be in alignment with a lower portion of the waste disposal chute **220** of the waste collection assembly **116** of the grill **100**. Unlike the air intake openings described above, the lower waste disposal opening **402** is not configured to draw air from the surrounding atmosphere into the cookbox **102**, but is instead configured to transport cooking waste passing through the waste disposal chute **220** into the disposable liner **218** and/or the waste bin **118** of the waste collection assembly **116** of the grill **100**, as further described above.

[0070] In the illustrated example of FIGS. **3** and **4**, the bottom wall **302** of the cookbox **102** has a width (e.g., measured between the right sidewall **308** and the left sidewall **310**) of approximately 25.0 inches, a depth (e.g., measured between the front wall **304** and the rear wall **306**) of approximately 13.9 inches, and an associated total area (e.g., calculated as width multiplied by depth) of approximately 347.5 square inches. The first air intake opening **312** formed in the bottom wall **302** has an area of approximately 39.0 square inches, and the second air intake opening **314** formed in the bottom wall **302** has an area of approximately 36.3 square inches. Thus, the first air intake opening **312** and the second air intake opening **314** collectively define an unoccupied area of 75.3 square inches. In the illustrated example of FIGS. **3** and **4**, the unoccupied area defined by the air intake openings accounts for approximately 21.7 percent of the total area of the bottom wall **302**. In other examples, the unoccupied area defined by the air intake openings can account for between 10.0 and 30.0 percent of the total area of the bottom wall **302**. Satisfaction of the above-

described range of the ratio of unoccupied area defined by the air intake openings to the total area of the bottom wall **302** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the ratio of unoccupied area defined by the air intake openings to the total area of the bottom wall **302** advantageously assists in minimizing any temperature variation across the flat top cooking surface of the griddle **104** during cooking operations performed thereon.

[0071] As shown in FIGS. **1-4**, the grill **100** further includes an example upper rim **132** defined by or associated with the cookbox **102**. In some examples, one or more portion(s) of the upper rim **132** of the grill **100** is/are formed by one or more portion(s) of the front wall **304**, the rear wall **306**, the right sidewall **308**, and/or the left sidewall **310** of the cookbox **102**. In some examples, one or more portion(s) of the upper rim **132** of the grill **100** is/are additionally or alternatively formed by one or more portion(s) of the control panel **108** and/or the frame **124** of the grill **100**. For example, as shown in FIGS. **1-4**, an example first portion **134** of the upper rim **132** (e.g., including a rear portion, a right side portion, and a left side portion of the upper rim **132**) is defined by the cookbox **102** of the grill **100**, and an example second portion **136** of the upper rim **132** is defined by the control panel **108** of the grill **100**. The cookbox **102**, the control panel **108**, and/or the frame **124** of the grill **100** include a plurality of example griddle docking openings **404** that are configured to receive and/or to otherwise engage corresponding ones of a plurality of griddle support members of the griddle **104**. As further described below, the griddle docking openings **404** and the griddle support members are configured to support the griddle **104** of the grill **100** on or above the cookbox **102** of the grill **100**, and/or on or above the upper rim **132** of the grill **100**.

[0072] FIG. **5** is a first perspective view showing the griddle **104** of the grill **100** of FIGS. **1** and **2** in isolation. FIG. **6** is a second perspective view of the griddle **104** of FIG. **5**. As shown in FIGS. **5** and **6**, the griddle **104** includes an example base **502**. The base **502** of the griddle **104** includes an example flat top cooking surface **504** and an example underside **602** located opposite the flat top cooking surface **504**. In the illustrated example of FIGS. **5** and **6**, the flat top cooking surface **504** is a continuous, substantially flat and/or substantially planar surface that is free of openings aside from the upper waste disposal opening **122** described herein. The flat top cooking surface **504** is configured to support a variety of liquid, semi-liquid, and/or solid food items during a variety of cooking processes that may be performed on the griddle **104**. In the illustrated example of FIGS. **5** and **6**, the flat top cooking surface **504** and/or, more generally, the base **502** of the griddle **104** has a generally rectangular profile. In other examples, the flat top cooking surface **504** and/or, more generally, the base **502** of the griddle **104** can instead have a non-rectangular profile (e.g., a circular profile, an oval profile, a triangular profile, a trapezoidal profile, etc.).

[0073] The griddle **104** further includes an example front lip **506**, and an example rear lip **508**, an example right side lip **510**, and an example left side lip **512**. The rear lip **508** of the griddle **104** is located opposite the front lip **506** of the griddle **104**. The right side lip **510** of the griddle **104** extends between the front lip **506** and the rear lip **508** of the griddle **104**. The left side lip **512** of the griddle **104** is located opposite the right side lip **510** of the griddle **104** and extends between the front lip **506** and the rear lip **508** of the griddle **104**. The front lip **506**, the rear lip **508**, the right side lip **510**, and the left side lip **512** of the griddle **104** extend upwardly from the base **502** of the griddle **104** to provide vertical boundaries configured to prevent food items from sliding off the flat top cooking surface **504** of the griddle **104** during one or more cooking operation(s) performed thereon.

[0074] As shown in FIGS. **5** and **6**, the upper waste disposal opening **122** formed in the base **502** of the griddle **104** extends completely through the base **502** (e.g., from the flat top cooking surface **504** of the base **502** through to the underside **602** of the base **502**). The upper waste disposal opening **122** facilitates the removal and/or disposal of liquid and/or solid cooking waste (e.g., grease, residue, particles, and/or other byproducts associated with a cooking operation) located on

the flat top cooking surface **504** of the griddle **104**, as further described above. In the illustrated example of FIGS. **5** and **6**, the upper waste disposal opening **122** is located proximate the front left corner of the base **502** of the griddle **104** (e.g., near the junction of the front lip **506** and the left side lip **512** of the griddle **104**). In other examples, the upper waste disposal opening **122** can instead be located proximate some other portion (e.g., the rear left corner, the front right corner, the rear right corner, etc.) of the base **502** of the griddle **104**.

[0075] The griddle **104** further includes a plurality of example griddle support members **604** (e.g., vertically oriented posts) coupled to and extending downwardly from the underside **602** of the base **502** of the griddle **104**. The griddle support members **604** are configured to support the underside **602** and/or, more generally, the base **502** of the griddle **104** on or above the cookbox **102** of the grill **100**, and/or on or above the upper rim **132** of the grill **100**. In this regard, respective ones of the griddle support members **604** of the griddle **104** are configured to be received by and/or to otherwise engage corresponding ones of the griddle docking openings **404** of the cookbox **102** of the grill **100**. In the illustrated example of FIGS. **5** and **6**, the griddle **104** includes a total of four griddle support members **604** configured to be received by and/or to otherwise engage a total of four griddle docking openings **404**. In other examples, the griddle **104** can instead include a different number (e.g., two, three, five, six, etc.) of griddle support members **604**, and the cookbox **102** can similarly include a corresponding different number (e.g., two, three, five, six, etc.) of griddle docking openings **404**.

[0076] The griddle **104** further includes a plurality of example anti-warping braces **606** coupled to and extending downwardly from the underside **602** of the base **502** of the griddle **104**. The anti-warping braces **606** are configured to support, strengthen, and/or brace the underside **602** and/or, more generally, the base **502** of the griddle **104** to limit and/or prevent any buckling and/or warping thereof. As shown in FIGS. **5** and **6**, the anti-warping braces **606** are arranged in a right-to-left orientation along the underside **602** of the base **502** of the griddle **104**. In other examples, the anti-warping braces **606** can instead be arranged in a different orientation (e.g., a front-to-rear orientation, a diagonal orientation, etc.) along the underside **602** of the base **502** of the griddle **104**. In the illustrated example of FIGS. **5** and **6**, the griddle **104** includes a total of three anti-warping braces **606**. In other examples, the griddle **104** can instead include a different number (e.g., one, two, four, etc.) of anti-warping braces **606**.

[0077] In the illustrated example of FIGS. **5** and **6**, the flat top cooking surface **504** and/or, more generally, the base **502** of the griddle **104** has a width (e.g., measured between the right side lip **510** to the left side lip **512**) of approximately 28.0 inches, a depth (e.g., measured between the front lip **506** and the rear lip **508**) of approximately 18.0 inches, and an associated total area (e.g., calculated as width multiplied by depth) of approximately 504.0 square inches. In other examples, the width, the depth, and/or the total area of the flat top cooking surface **504** can differ from the above-described dimensions.

[0078] FIG. **7** is a perspective view showing the first burner tube **202** of the grill **100** of FIGS. **1** and **2** in isolation. FIG. **8** is a top view of the first burner tube **202** of FIG. **7**. FIG. **9** is a bottom view of the first burner tube **202** of FIGS. **7** and **8**. FIG. **10** is a right side view of the first burner tube **202** of FIGS. **7-9**. FIG. **11** is a cross-sectional view of the first burner tube **202** of FIGS. **7-10** taken along section A-A of FIG. **8**. The construction of the second burner tube **204** and the third burner tube **206** of the grill **100** of FIGS. **1** and **2** is substantially identical to that of the first burner tube **202** as described herein in connection with FIGS. **7-11**.

[0079] As shown in FIGS. **7-11**, the first burner tube **202** includes an example first end **702** (e.g., a front end), an example second end **704** (e.g., a rear end), an example outer wall **706** extending between the first end **702** and the second end **704**, and an example central axis **708** extending between the first end **702** and the second end **704**. The first end **702** of the first burner tube **202** is partially open, with the first end **702** being configured to receive an outlet of a control valve (e.g., the outlet of the first control valve **212**) located proximate the front wall **304** of the cookbox **102**.

The second end **704** of the first burner tube **202** is closed, with the second end **704** being configured to be coupled to a mounting flange located proximate the rear wall **306** of the cookbox **102**. The first burner tube **202** is accordingly configured to be arranged in a front-to-rear orientation when disposed in the cookbox **102**. In the illustrated example of FIGS. **7-11**, the first burner tube **202** is structured as a linear burner tube having a circular cross-sectional profile. In other example, the first burner tube **202** can instead be structured as a linear burner tube having a different cross-sectional profile (e.g., a rectangular cross-sectional profile, an oval-shaped cross-sectional profile, a triangular cross-sectional profile, a trapezoidal cross-sectional profile, etc.). In still other examples, the first burner tube **202** can instead be structured as a non-linear burner tube having one or more bend(s) and/or curve(s) formed therein.

[0080] The first burner tube **202** of FIGS. **7-11** includes a plurality of ports formed in and extending through the outer wall **706** of the first burner tube **202**. As shown in FIGS. **7-11**, the plurality of ports includes an example first row of ports **710**, an example second row of ports **712**, an example third row of ports **714**, and an example fourth row of ports **716**. The first row of ports **710** is arranged parallel to the central axis **708**. The second row of ports **712** is also arranged parallel to the central axis **708**, with the second row of ports **712** being spaced apart from (e.g., angularly displaced from) the first row of ports **710** about the perimeter (e.g., about the circumference) of the outer wall **706** of the first burner tube **202**. The third row of ports **714** is arranged perpendicular to the central axis **708**. The fourth row of ports **716** is also arranged perpendicular to the central axis **708**, with the fourth row of ports **716** being spaced apart from (e.g., longitudinally displaced from) the third row of ports **714** along central axis **708** of the first burner tube **202**. In the illustrated example of FIGS. **7-11**, the third row of ports **714** is located proximate the first end **702** (e.g., the front end) of the first burner tube **202**, and the fourth row of ports **716** is located proximate the second end **704** (e.g., the rear end) of the first burner tube **202**. The first burner tube **202** can include additional ports and/or additional rows of ports relative to those described above. For example, as shown in FIGS. **7-11**, the first burner tube **202** further includes an example fifth row of ports **718** arranged perpendicular to the central axis **708**, with the fifth row of ports **718** being spaced apart from and proximate to the third row of ports **714**.

[0081] As shown in FIGS. **7-11**, the outer wall **706** of the first burner tube **202** has a circular cross-sectional profile. The second row of ports **712** is angularly displaced from the first row of ports **710** about the circumference of the outer wall **706** by an example angle **1102** measuring approximately 120.0 degrees. In other examples, the angle **1102** at which the second row of ports **712** is angularly displaced from the first row of ports **710** about the circumference of the outer wall **706** can be between 90.0 and 150.0 degrees. Satisfaction of the above-described range of the angle **1102** at which the second row of ports **712** is angularly displaced from the first row of ports **710** about the circumference of the outer wall **706** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the angle **1102** at which the second row of ports **712** is angularly displaced from the first row of ports **710** about the circumference of the outer wall **706** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104** during cooking operations performed thereon.

[0082] The first burner tube **202** of FIGS. **7-11** is further configured such that the third row of ports **714** of the first burner tube **202** is spaced apart from the first end **702** (e.g., the front end) of the first burner tube **202** by an example distance **802** of approximately 3.8 inches. In other examples, the distance **802** between the third row of ports **714** and the first end **702** of the first burner tube **202** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **802** between the third row of ports **714** and the first end **702** of the first burner tube **202** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this

regard, satisfaction of the above-described range of the distance **802** between the third row of ports **714** and the first end **702** of the first burner tube **202** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104** during cooking operations performed thereon.

[0083] The first burner tube **202** of FIGS. **7-11** is further configured such that the fourth row of ports **716** of the first burner tube **202** is spaced apart from the second end **704** (e.g., the rear end) of the first burner tube **202** by an example distance **804** of approximately 2.8 inches. In other examples, the distance **804** between the fourth row of ports **716** and the second end **704** of the first burner tube **202** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **804** between the fourth row of ports **716** and the second end **704** of the first burner tube **202** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the distance **804** between the fourth row of ports **716** and the second end **704** of the first burner tube **202** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104** during cooking operations performed thereon.

[0084] In the illustrated example of FIGS. **7-11**, the first burner tube **202** has a length of approximately 18.0 inches measured along the central axis **708** from the first end **702** (e.g., the front end) to the second end **704** (e.g., the rear end) of the first burner tube **202**. In other examples, the first burner tube **202** can instead have a length that is substantially greater than or substantially less than 18.0 inches. In the illustrated example of FIGS. **7-11**, the outer wall **706** of the first burner tube **202** has a diameter of approximately 1.0 inches measured across the circular cross-sectional profile of the outer wall **706**. In other examples, the first burner tube **202** can instead have a diameter that is substantially greater than or substantially less than 1.0 inches. In the illustrated example of FIGS. **7-11**, each one of the ports formed in and extending through the outer wall **706** of the first burner tube **202** has a diameter of approximately 0.07 inches measured across the circular outlet opening of the port. In other examples, one or more of the ports formed in and extending through the outer wall **706** of the first burner tube **202** can instead have a diameter that is substantially greater than or substantially less than 0.07 inches.

[0085] FIG. **12** is a front view showing the grill **100** of FIGS. **1** and **2**, with the frame **124**, the lid **128**, and certain other structural features of the grill **100** omitted for enhanced viewability. FIG. **13** is a right side view of the grill **100** of FIG. **12**. FIG. **14** is a top view of the grill **100** of FIGS. **12** and **13**. FIG. **15** is a cross-sectional view of the grill **100** of FIGS. **12-14** taken along section B-B of FIG. **14**. FIG. **16** is a cross-sectional view of the grill **100** of FIGS. **12-15** taken along section C-C of FIG. **14**. FIG. **17** is a cross-sectional view of the grill **100** of FIGS. **12-16** taken along section D-D of FIG. **12**.

[0086] As shown in FIGS. **12-17**, the griddle **104** of the grill **100** is disposed on or above the cookbox **102** of the grill **100**. More specifically, the griddle **104** is removably positioned and/or docked on top of the cookbox **102** via engagement of the griddle support members **604** of the griddle **104** and the griddle docking openings **404** associated with the cookbox **102**. As further shown in FIGS. **12-17**, the underside **602** of the griddle **104** is located above and spaced apart from the upper rim **132** of the cookbox **102** by an example vertical gap **1202** that is configured to vent heated air generated within the cookbox **102** (e.g., heated air generated via the first burner tube **202**, the second burner tube **204**, and/or the third burner tube **206**). In addition to extending between the upper rim **132** of the cookbox **102** and the underside **602** of the griddle **104**, the vertical gap **1202** further extends completely or substantially around a perimeter of the upper rim **132** of the cookbox **102**, and completely or substantially around a perimeter of the underside **602** of the griddle **104**. Configuring the vertical gap **1202** in this manner advantageously facilitates even and/or uniform venting of heat from the cookbox **102** (e.g., along each of the front wall **304**, the rear wall **306**, the right sidewall **308**, and the left sidewall **310** of the cookbox **102**) relative to the

griddle **104** (e.g., along the front lip **506**, the rear lip **508**, the right side lip **510**, and the left side lip **512** of the griddle **104**).

[0087] In the illustrated example of FIGS. **12-17**, the vertical gap **1202** has a height of approximately 0.70 inches. In other examples, the vertical gap **1202** can have a height between 0.25 and 1.50 inches. Satisfaction of the above-described range of the height of the vertical gap **1202** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the height of the vertical gap **1202** advantageously assists in minimizing any temperature variation across the flat top cooking surface of the griddle **104** during cooking operations performed thereon.

[0088] As shown in FIGS. **12-17**, the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) of the grill **100** are disposed within the cookbox **102** of the grill **100** at a location between the bottom wall **302** of the cookbox **102** and the underside **602** of the griddle **104**. More specifically, the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) of the grill **100** are disposed within the cookbox **102** of the grill **100** at a location between the bottom wall **302** of the cookbox **102** and the upper rim **132** of the cookbox **102**, with respective ones of the burner tubes being laterally spaced apart from one another and arranged in a front-to-rear orientation. As shown in FIGS. **12-17**, the first burner tube **202** is laterally positioned to the right side of the first air intake opening **312** formed in the bottom wall **302** of the cookbox **102**. The second burner tube **204** is laterally positioned to the left side of the first air intake opening **312** formed in the bottom wall **302** of the cookbox **102** and to the right side of the second air intake opening **314** formed in the bottom wall **302** of the cookbox **102**. The second burner tube **204** is accordingly located between the first air intake opening **312** and the second air intake opening **314** formed in the bottom wall **302** of the cookbox **102**, such that the second burner tube **204** is laterally aligned with the solid portion **316** of the bottom wall **302** of the cookbox **102**. The third burner tube **206** is laterally positioned to the left side of the second air intake opening **314** formed in the bottom wall **302** of the cookbox **102**.

[0089] In the illustrated example of FIGS. **12-17**, each of the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) is vertically positioned above the bottom wall **302** of the cookbox **102** by an example distance **1502** of approximately 2.5 inches measured vertically from the bottom wall **302** of the cookbox **102** to the central axis **708** of the respective burner tube. In other examples, the distance **1502** at which each of the burner tubes is vertically positioned above the bottom wall **302** of the cookbox **102** can be between 1.5 and 3.5 inches. Satisfaction of the above-described range of the distance **1502** at which each of the burner tubes is vertically positioned above the bottom wall **302** of the cookbox **102** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the distance **1502** at which each of the burner tubes is vertically positioned above the bottom wall **302** of the cookbox **102** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104** during cooking operations performed thereon.

[0090] In the illustrated example of FIGS. **12-17**, each of the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) is vertically positioned below the upper rim **132** of the cookbox **102** by an example distance **1504** of approximately 3.1 inches measured vertically from the upper rim **132** of the cookbox **102** to the central axis **708** of the respective burner tube. In other examples, the distance **1504** at which each of the burner tubes is vertically positioned below the upper rim **132** of the cookbox **102** can be between 2.0 and 4.0 inches. Satisfaction of the above-described range of the distance **1504** at which each of the burner tubes is vertically positioned below the upper rim **132** of the cookbox **102** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat

distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the distance **1504** at which each of the burner tubes is vertically positioned below the upper rim **132** of the cookbox **102** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104** during cooking operations performed thereon.

[0091] In the illustrated example of FIGS. **12-17**, each of the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) is vertically positioned below the underside **602** of the griddle **104** by an example distance **1506** of approximately 3.8 inches measured vertically from the underside **602** of the griddle **104** to the central axis **708** of the respective burner tube. In other examples, the distance **1506** at which each of the burner tubes is vertically positioned below the underside **602** of the griddle **104** can be between 3.0 and 5.0 inches. Satisfaction of the above-described range of the distance **1506** at which each of the burner tubes is vertically positioned below the underside **602** of the griddle **104** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the distance **1506** at which each of the burner tubes is vertically positioned below the underside **602** of the griddle **104** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104** during cooking operations performed thereon.

[0092] In the illustrated example of FIGS. **12-17**, the third row of ports **714** formed on the outer wall **706** of each of the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) is positioned inwardly from the front wall **304** of the cookbox **102** by an example distance **1602** of approximately 3.0 inches measured depthwise (e.g., along the central axis **708** of the respective burner tube) from the front wall **304** of the cookbox **102** to the third row of ports **714** of the respective burner tube. In other examples, the distance **1602** at which the third row of ports **714** of the respective burner tube is positioned inwardly from the front wall **304** of the cookbox **102** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **1602** at which the third row of ports **714** of the respective burner tube is positioned inwardly from the front wall **304** of the cookbox **102** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the distance **1602** at which the third row of ports **714** of the respective burner tube is positioned inwardly from the front wall **304** of the cookbox **102** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104** during cooking operations performed thereon.

[0093] In the illustrated example of FIGS. **12-17**, the fourth row of ports **716** formed on the outer wall **706** of each of the burner tubes (e.g., the first burner tube **202**, the second burner tube **204**, and the third burner tube **206**) is positioned inwardly from the rear wall **306** of the cookbox **102** by an example distance **1604** of approximately 2.0 inches measured depthwise (e.g., along the central axis **708** of the respective burner tube) from the rear wall **306** of the cookbox **102** to the fourth row of ports **716** of the respective burner tube. In other examples, the distance **1604** at which the fourth row of ports **716** of the respective burner tube is positioned inwardly from the rear wall **306** of the cookbox **102** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **1604** at which the fourth row of ports **716** of the respective burner tube is positioned inwardly from the rear wall **306** of the cookbox **102** advantageously improves the operating efficiency of the cooking engine of the grill **100**, and also improves the heat distribution properties associated with the griddle **104** of the grill **100**. In this regard, satisfaction of the above-described range of the distance **1604** at which the fourth row of ports **716** of the respective burner tube is positioned inwardly from the rear wall **306** of the cookbox **102** advantageously assists in minimizing any temperature variation across the flat top cooking surface **504** of the griddle **104**

during cooking operations performed thereon.

[0094] In the illustrated example of FIGS. 12-17, the third row of ports 714 formed on the outer wall 706 of each of the burner tubes (e.g., the first burner tube 202, the second burner tube 204, and the third burner tube 206) is positioned inwardly from the front lip 506 of the griddle 104 by an example distance 1606 of approximately 3.5 inches measured depthwise (e.g., along the central axis 708 of the respective burner tube) from the front lip 506 of the griddle 104 to the third row of ports 714 of the respective burner tube. In other examples, the distance 1606 at which the third row of ports 714 of the respective burner tube is positioned inwardly from the front lip 506 of the griddle 104 can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance 1606 at which the third row of ports 714 of the respective burner tube is positioned inwardly from the front lip 506 of the griddle 104 advantageously improves the operating efficiency of the cooking engine of the grill 100, and also improves the heat distribution properties associated with the griddle 104 of the grill 100. In this regard, satisfaction of the above-described range of the distance 1606 at which the third row of ports 714 of the respective burner tube is positioned inwardly from the front lip 506 of the griddle 104 advantageously assists in minimizing any temperature variation across the flat top cooking surface 504 of the griddle 104 during cooking operations performed thereon.

[0095] In the illustrated example of FIGS. 12-17, the fourth row of ports 716 formed on the outer wall 706 of each of the burner tubes (e.g., the first burner tube 202, the second burner tube 204, and the third burner tube 206) is positioned inwardly from the rear lip 508 of the griddle 104 by an example distance 1608 of approximately 3.1 inches measured depthwise (e.g., along the central axis 708 of the respective burner tube) from the rear lip 508 of the griddle 104 to the fourth row of ports 716 of the respective burner tube. In other examples, the distance 1608 at which the fourth row of ports 716 of the respective burner tube is positioned inwardly from the rear lip 508 of the griddle 104 can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance 1608 at which the fourth row of ports 716 of the respective burner tube is positioned inwardly from the rear lip 508 of the griddle 104 advantageously improves the operating efficiency of the cooking engine of the grill 100, and also improves the heat distribution properties associated with the griddle 104 of the grill 100. In this regard, satisfaction of the above-described range of the distance 1608 at which the fourth row of ports 716 of the respective burner tube is positioned inwardly from the rear lip 508 of the griddle 104 of the cookbox 102 advantageously assists in minimizing any temperature variation across the flat top cooking surface 504 of the griddle 104 during cooking operations performed thereon.

[0096] The grill 100 of FIGS. 1-17 is configured for optimum efficiency and optimum heat distribution during cooking operations performed on the flat top cooking surface 504 of the griddle 104. More specifically, the grill 100 is configured to minimize any temperature variation across the flat top cooking surface 504 of the griddle 104 during cooking operations performed thereon. As shown in FIG. 14, the flat top cooking surface 504 of the griddle 104 includes an example central portion 1402 that is bounded and/or circumscribed by an example peripheral portion 1404. In the illustrated example, the peripheral portion 1404 of the flat top cooking surface 504 extends approximately three inches inwardly from each of the four surrounding lips of the griddle 104 (e.g., the front lip 506, the rear lip 508, the right side lip 510, and the left side lip 512). The grill 100 of FIGS. 1-17 is configured such that the average temperature within the central portion 1402 of the flat top cooking surface 504 of the griddle 104 is approximately 375 degrees Fahrenheit (375° F.) when the burner tubes (e.g., the first burner tube 202, the second burner tube 204, and the third burner tube 206) of the grill 100 are adjusted to a LOW output setting, approximately 420 degrees Fahrenheit (420° F.) when the burner tubes of the grill 100 are adjusted to a MEDIUM output setting, and approximately 560 degrees Fahrenheit (560° F.) when the burner tubes of the grill 100 are adjusted to a HI output setting. As a result of implementing the innovative heat distribution features described above, the grill 100 of FIGS. 1-17 is advantageously configured such that the

average temperature variation between the central portion **1402** and the peripheral portion **1404** of the flat top cooking surface **504** of the griddle **104** is approximately 60 degrees Fahrenheit (60° F.) or less when the burner tubes of the grill **100** are adjusted to a LOW output setting, approximately 90 degrees Fahrenheit (90° F.) or less when the burner tubes of the grill **100** are adjusted to a MEDIUM output setting, and approximately 140 degrees Fahrenheit (140° F.) or less when the burner tubes of the grill **100** are adjusted to a HI output setting. The above-described temperature variations are substantially less than those found in many commercially available flat top gas grills, some of which produce flat top cooking surface temperature variations exceeding two hundred degrees Fahrenheit (200° F.) when the burner tubes of the flat top gas grill are adjusted to a common output setting.

[0097] FIG. **18** is a perspective view of another example grill **1800** constructed in accordance with the teachings of this disclosure. FIG. **19** is an exploded view of the grill **1800** of FIG. **18**. The grill **1800** of FIGS. **18** and **19** is structured as a flat top gas grill. In this regard, the grill **1800** of FIGS. **18** and **19** includes an example cookbox **1802** and an example griddle **1804**. The griddle **1804** of the grill **1800** is configured to be disposed on or above the cookbox **1802** of the grill **1800**. The cookbox **1802** of the grill **1800** is further described below in connection with FIGS. **20** and **21**. The griddle **1804** of the grill **1800** is further described below in connection with FIGS. **22** and **23**.

[0098] The grill **1800** of FIGS. **18** and **19** further includes a plurality of burner tubes configured to be disposed within the cookbox **1802** at a location between a bottom wall of the cookbox **1802** and an upper rim defined by or associated with the cookbox **1802**, and/or at a location between the bottom wall of the cookbox **1802** and an underside of the griddle **1804**. As shown in FIG. **19**, the plurality of burner tubes includes an example first burner tube **1902**, an example second burner tube **1904**, an example third burner tube **1906**, and an example fourth burner tube **1908** (e.g., a total of four burner tubes) that are configured to be laterally spaced apart from one another and arranged in a front-to-rear orientation within the cookbox **1802**, with the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908** being of a substantially identical construction relative to one another. In other examples, the plurality of burner tubes can instead include a different number (e.g., two, three, five, etc.) of burner tubes, and the construction of one or more of the burner tubes may differ from that of the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and/or the fourth burner tube **1908** shown in FIG. **19**. The first burner tube **1902** of the grill **1800** is further described below in connection with FIGS. **24-28**.

[0099] In the illustrated example of FIGS. **18** and **19**, the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) form part of a gas train that further includes an example fuel source **1806**, an example regulator assembly **1910**, an example manifold **1912**, and a plurality of control valves corresponding in number to the plurality of burner tubes. In this regard, the plurality of control valves of the grill **1800** as shown in FIG. **19** includes an example first control valve **1914** associated with the first burner tube **1902**, an example second control valve **1916** associated with the second burner tube **1904**, an example third control valve **1918** associated with the third burner tube **1906**, and an example fourth control valve **1920** associated with the fourth burner tube **1908**.

[0100] In the illustrated example of FIG. **18**, the fuel source **1806** is implemented as a fuel tank (e.g., a propane tank) containing combustible gas. In other examples, the fuel source **1806** can instead be implemented as a piped (e.g., household) natural gas line that provides an accessible flow of combustible gas. The regulator assembly **1910** is operatively positioned between the fuel source **1806** and the manifold **1912** such that a supply of combustible gas provided via the fuel source **1806** flows through the regulator assembly **1910** and into the manifold **1912**. The first control valve **1914** is operatively positioned between the manifold **1912** and the first burner tube **1902** such that combustible gas received at the manifold **1912** can be selectively supplied to the first burner tube **1902** via the first control valve **1914**. The second control valve **1916** is operatively

positioned between the manifold **1912** and the second burner tube **1904** such that combustible gas received at the manifold **1912** can be selectively supplied to the second burner tube **1904** via the second control valve **1916**. The third control valve **1918** is operatively positioned between the manifold **1912** and the third burner tube **1906** such that combustible gas received at the manifold **1912** can be selectively supplied to the third burner tube **1906** via the third control valve **1918**. The fourth control valve **1920** is operatively positioned between the manifold **1912** and the third burner tube **1906** such that combustible gas received at the manifold **1912** can be selectively supplied to the third burner tube **1906** via the third control valve **1918**.

[0101] In the illustrated example of FIGS. **18** and **19**, the manifold **1912**, the first control valve **1914**, the second control valve **1916**, the third control valve **1918**, and the fourth control valve **1920** are at least partially covered and/or concealed by an example control panel **1808** that is coupled to and/or located along the front of the cookbox **1802**. The grill **1800** further includes a plurality of control knobs mounted and/or located along the front face of the control panel **1808**, with the plurality of control knobs corresponding in number to the plurality of control valves and/or the number of burner tubes. In this regard, the plurality of control valves of the grill **1800** as shown in FIGS. **18** and **19** includes an example first control knob **1810** associated with the first control valve **1914** and/or the first burner tube **1902**, an example second control knob **1812** associated with the second control valve **1916** and/or the second burner tube **1904**, an example third control knob **1814** associated with the third control valve **1918** and/or the third burner tube **1906**, and an example fourth control knob **1816** associated with the fourth control valve **1920** and/or the fourth burner tube **1908**. Each control knob is mechanically coupled to its corresponding control valve such that movement (e.g., rotation) of the control knob changes the extent to which an adjustable flow control member of the corresponding control valve enables combustible gas to flow through the corresponding control valve into the corresponding burner tube. For example, the first control knob **1810** is mechanically coupled to the first control valve **1914** such that movement (e.g., rotation) of the first control knob **1810** changes the extent to which an adjustable flow control member of the first control valve **1914** enables combustible gas to flow through the first control valve **1914** into the first burner tube **1902**.

[0102] The grill **1800** of FIGS. **18** and **19** further includes an example waste collection assembly **1818** configured to collect and facilitate the removal of cooking waste. In the illustrated example of FIGS. **18** and **19**, the waste collection assembly **1818** includes an example waste bin **1820** that is suspended from and slidably coupled to a pair of example support rails **1822**, with the support rails **1822** being coupled to an underside of the cookbox **1802**. The waste bin **1820** is moveable (e.g., slidable) along the support rails **1822** between a closed position in which a substantial portion of the waste bin **1820** is covered by the underside of the cookbox **1802** (e.g., as shown in FIG. **18**), and an open position in which a substantial portion of the waste bin **1820** is located forward of and not covered by the underside of the cookbox **1802**. The waste bin **1820** is configured to hold and/or contain an example disposable liner **1922**.

[0103] When the waste bin **1820** is in the closed position, the waste bin **1820** and/or the disposable liner **1922** is/are positioned below a lower waste disposal opening formed in and extending through a bottom wall of the cookbox **1802**. The waste collection assembly **1818** further includes an example waste disposal chute **1924** disposed within the cookbox **1802**, and an example upper waste disposal opening **1824** formed in and extending through the griddle **1804**. The waste disposal chute **1924** is operatively positioned between the upper waste disposal opening **1824** and the lower waste disposal opening such that liquid and/or solid cooking waste (e.g., grease, residue, particles, and/or other byproducts associated with a cooking operation) located on a flat top cooking surface of the griddle **1804** can be fed into the upper waste disposal opening **1824**, with the cooking waste thereafter passing from the upper waste disposal opening **1824** through the waste disposal chute **1924** and through the lower waste disposal opening. The waste bin **1820** and/or the disposable liner **1922** collect(s) cooking waste that passes through the waste disposal chute **1924** and the lower

waste disposal opening when the waste bin **1820** is in the closed position. Moving the waste bin **1820** from the closed position into the open position facilitates removal of the disposable liner **1922** and/or the cooking waste contained therein.

[0104] The grill **1800** of FIGS. **18** and **19** further includes an example frame **1826**. In the illustrated example of FIGS. **18** and **19**, the frame **1826** includes one or more example cookbox support member(s) **1828** (e.g., one or more vertically oriented leg(s)) that are configured to support the cookbox **1802** above an underlying ground surface. The cookbox support member(s) **1828** and/or, more generally, the frame **1826** can be configured from any number and any type of structural components arranged in any manner that facilitates supporting the cookbox **1802** above an underlying ground surface when the grill **1800** is in use. For example, while each of the one or more cookbox support member(s) **1828** shown in FIGS. **18** and **19** is fixed relative to the cookbox **1802**, in other examples the frame **1826** can include one or more foldable, slidable, and/or telescoping cookbox support member(s) **1828** that facilitate collapsing and/or otherwise modifying the frame **1826** of the grill **1800** when the grill **1800** is not in use. The grill **1800** of FIGS. **18** and **19** can further include one or more side table(s) coupled to the frame **1826** and/or to the cookbox **1802** of the grill **1800**.

[0105] The grill **1800** of FIGS. **18** and **19** further includes an example lid **1830** configured to cover and/or enclose the griddle **1804** of the grill **1800** when the lid **1830** is in a closed position. The lid **1830** is movable relative to the griddle **1804** and/or the cookbox **1802** between a closed position and an open position in which the flat top cooking surface of the griddle **1804** is exposed (e.g., as shown in FIG. **18**). In the illustrated example of FIGS. **18** and **19**, the lid **1830** can be pivoted relative to the cookbox **1802**, the griddle **1804**, and/or the frame **1826** via one or more hinge(s) that mechanically couple the lid **1830** to the cookbox **1802** and/or the frame **1826**. In other example, the lid **1830** can instead be removably positioned on the griddle **1804** and/or the cookbox **1802** without there being any direct mechanical coupling between the lid **1830** on the one hand and the cookbox **1802**, the griddle **1804**, and/or the frame **1826** on the other hand. Movement of the lid **1830** of the grill **1800** between the closed position and the open position can be facilitated via user interaction with an example handle **1832** that is coupled to the lid **1830**.

[0106] FIG. **20** is a perspective view showing the cookbox **1802** of the grill **1800** of FIGS. **18** and **19** in isolation. FIG. **21** is a top view of the cookbox **1802** of FIG. **20**. As shown in FIGS. **20** and **21**, the cookbox **1802** includes an example bottom wall **2002**, an example front wall **2004**, an example rear wall **2006**, an example right sidewall **2008**, and an example left sidewall **2010**. The rear wall **2006** of the cookbox **1802** is located opposite the front wall **2004** of the cookbox **1802**. The right sidewall **2008** of the cookbox **1802** extends between the front wall **2004** and the rear wall **2006** of the cookbox **1802**. The left sidewall **2010** of the cookbox **1802** is located opposite the right sidewall **2008** of the cookbox **1802** and extends between the front wall **2004** and the rear wall **2006** of the cookbox **1802**. As shown in FIGS. **20** and **21**, the front wall **2004**, the rear wall **2006**, the right sidewall **2008**, and the left sidewall **2010** of the cookbox **1802** extend upwardly from the bottom wall **2002** of the cookbox **1802** to define a cavity in which one or more structural component(s) (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) of the grill **1800** can be disposed.

[0107] The cookbox **1802** further includes a plurality of air intake openings formed in and extending through the bottom wall **2002** of the cookbox **1802**. The air intake openings are individually and collectively configured to draw air from the surrounding atmosphere into the cookbox **1802**. As shown in FIGS. **20** and **21**, the plurality of air intake openings includes an example first air intake opening **2102**, an example second air intake opening **2104**, an example third air intake opening **2106**, an example fourth air intake opening **2108**, an example fifth air intake opening **2110**, an example sixth air intake opening **2112**, and an example seventh air intake opening **2114**. The first air intake opening **2102** and the second air intake opening **2104** are collectively laterally spaced apart from the third air intake opening **2016**, the fourth air intake

opening **2108**, and the fifth air intake opening **2110**. The third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** are collectively laterally spaced apart from the sixth air intake opening **2112** and the seventh air intake opening **2114**. In this regard, the bottom wall **2002** of the cookbox **1802** further includes an example first solid portion **2116** and an example second solid portion **2118** that are laterally spaced apart from one another, and that respectively extend between the front wall **2004** and the rear wall **2006** of the cookbox **1802**. The first solid portion **2116** further extends between the first air intake opening **2102** and the second air intake opening **2104** on the one hand, and the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** on the other hand. The second solid portion **2118** further extends between the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** on the one hand, and the sixth air intake opening **2112** and the seventh air intake opening **2114** on the other hand. In the illustrated example of FIGS. **20** and **21**, the first solid portion **2116** and the second solid portion **2118** are free of any unplugged and/or uncovered opening(s) that extend through the bottom wall **2002**.

[0108] The first air intake opening **2102**, the second air intake opening **2104**, the third air intake opening **2106**, the fourth air intake opening **2108**, the fifth air intake opening **2110**, and the first solid portion **2116** of the bottom wall **2002** of the cookbox **1802** are individually and collectively configured such that a burner tube (e.g., the second burner tube **1904**) of the grill **1800** is laterally aligned with the first solid portion **2116**, and such that said burner tube (e.g., the second burner tube **1904**) is laterally positioned between the first air intake opening **2102** and the second air intake opening **2104** on the one hand and the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** on the other hand. The first air intake opening **2102**, the second air intake opening **2104**, the third air intake opening **2106**, the fourth air intake opening **2108**, the fifth air intake opening **2110**, and the first solid portion **2116** of the bottom wall **2002** of the cookbox **1802** are further individually and collectively configured such that the first air intake opening **2102** and the second air intake opening **2104** are laterally positioned between one set of neighboring burner tubes (e.g., the first burner tube **1902** and the second burner tube **1904**) of the grill **1800**, and such that the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** are laterally positioned between another set of neighboring burner tubes (e.g., the second burner tube **1904** and the third burner tube **1906**) of the grill **1800**.

[0109] The third air intake opening, **2106**, fourth air intake opening **2108**, the fifth air intake opening **2110**, the sixth air intake opening **2112**, the seventh air intake opening **2114**, and the second solid portion **2118** of the bottom wall **2002** of the cookbox **1802** are individually and collectively configured such that a burner tube (e.g., the third burner tube **1906**) of the grill **1800** is laterally aligned with the second solid portion **2118**, and such that said burner tube (e.g., the third burner tube **1906**) is laterally positioned between the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** on the one hand and the sixth air intake opening **2112** and the seventh air intake opening **2114** on the other hand. The third air intake opening **2106**, the fourth air intake opening **2108**, the fifth air intake opening **2110**, the sixth air intake opening **2112**, the seventh air intake opening **2114**, and the second solid portion **2118** of the bottom wall **2002** of the cookbox **1802** are further individually and collectively configured such that the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** are laterally positioned between one set of neighboring burner tubes (e.g., the second burner tube **1904** and the third burner tube **1906**) of the grill **1800**, and such that the sixth air intake opening **2112** and the seventh air intake opening **2114** are laterally positioned between another set of neighboring burner tubes (e.g., the third burner tube **1906** and the fourth burner tube **1908**) of the grill **1800**.

[0110] As shown in FIGS. **20** and **21**, the first air intake opening **2102** and the second air intake opening **2104** formed in the bottom wall **2002** of the cookbox **1802** are spaced apart from one another along a depthwise direction (e.g., front-to-rear) of the cookbox **1802**. The first air intake

opening **2102** and the second air intake opening **2104** are generally aligned with one another along the depthwise direction, thereby forming a first depthwise row of air intake openings. The third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** formed in the bottom wall **2002** of the cookbox **1802** are spaced apart from one another along a depthwise direction (e.g., front-to-rear) of the cookbox **1802**. The third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** are generally aligned with one another along the depthwise direction, thereby forming a second depthwise row of air intake openings that is laterally spaced apart from the first depthwise row of air intake openings. The sixth air intake opening **2112** and the seventh air intake opening **2114** formed in the bottom wall **2002** of the cookbox **1802** are spaced apart from one another along a depthwise direction (e.g., front-to-rear) of the cookbox **1802**. The sixth air intake opening **2112** and the seventh air intake opening **2114** are generally aligned with one another along the depthwise direction, thereby forming a third depthwise row of air intake openings that is laterally spaced apart from both the second depthwise row of air intake openings and the first depth row of air intake openings.

[0111] In the illustrated example of FIGS. **20** and **21**, the plurality of air intake openings includes a total of seven air intake openings (e.g., the first air intake opening **2102**, the second air intake opening **2104**, the third air intake opening **2106**, the fourth air intake opening **2108**, the fifth air intake opening **2110**, the sixth air intake opening **2112**, and the seventh air intake opening **2114**). In other examples, the plurality of air intake openings can instead include a different number (e.g., two, three, four, five, six, eight, nine, etc.) of air intake openings. In the illustrated example of FIGS. **20** and **21**, the first air intake opening **2102**, the second air intake opening **2104**, the third air intake opening **2106**, the fifth air intake opening **2110**, the sixth air intake opening **2112**, and the seventh air intake opening **2114** each have a generally rectangular shape, and the fourth air intake opening **2108** has a generally trapezoidal shape. In other examples, the first air intake opening **2102**, the second air intake opening **2104**, the third air intake opening **2106**, the fourth air intake opening **2108**, the fifth air intake opening **2110**, the sixth air intake opening **2112**, and/or the seventh air intake opening **2114** can instead have a different shape (e.g., a circular shape, an oval shape, a triangular shape, etc.).

[0112] In addition to the air intake openings illustrated in FIGS. **20** and **21**, the cookbox **1802** further includes an example lower waste disposal opening **2120** formed in and extending through the bottom wall **2002** of the cookbox **1802**. The lower waste disposal opening **2120** is configured to receive and/or to otherwise be in alignment with a lower portion of the waste disposal chute **1924** of the waste collection assembly **1818** of the grill **1800**. Unlike the air intake openings described above, the lower waste disposal opening **2120** is not configured to draw air from the surrounding atmosphere into the cookbox **1802**, but is instead configured to transport cooking waste passing through the waste disposal chute **1924** into the disposable liner **1922** and/or the waste bin **1820** of the waste collection assembly **1818** of the grill **1800**, as further described above.

[0113] In the illustrated example of FIGS. **20** and **21**, the bottom wall **2002** of the cookbox **1802** has a width (e.g., measured between the right sidewall **2008** and the left sidewall **2010**) of approximately 33.0 inches, a depth (e.g., measured between the front wall **2004** and the rear wall **2006**) of approximately 17.1 inches, and an associated total area (e.g., calculated as width multiplied by depth) of approximately 564.3 square inches. The first air intake opening **2102** formed in the bottom wall **2002** has an area of approximately 7.8 square inches. The second air intake opening **2104** formed in the bottom wall **2002** has an area of approximately 7.8 square inches. The third air intake opening **2106** formed in the bottom wall **2002** has an area of approximately 15.5 square inches. The fourth air intake opening **2108** formed in the bottom wall **2002** has an area of approximately 9.1 square inches. The fifth air intake opening **2110** formed in the bottom wall **2002** has an area of approximately 15.5 square inches. The sixth air intake opening **2112** formed in the bottom wall **2002** has an area of approximately 7.8 square inches. The seventh air intake opening **2114** formed in the bottom wall **2002** has an area of approximately 7.8 square

inches. Thus, the first air intake opening **2102**, the second air intake opening **2104**, the third air intake opening **2106**, the fourth air intake opening **2108**, the fifth air intake opening **2110**, the sixth air intake opening **2112**, and the seventh air intake opening **2114** collectively define an unoccupied area of 71.3 square inches. In the illustrated example of FIGS. **20** and **21**, the unoccupied area defined by the air intake openings accounts for approximately 12.6 percent of the total area of the bottom wall **2002**. In other examples, the unoccupied area defined by the air intake openings can account for between 10.0 and 30.0 percent of the total area of the bottom wall **2002**. Satisfaction of the above-described range of the ratio of unoccupied area defined by the air intake openings to the total area of the bottom wall **2002** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the ratio of unoccupied area defined by the air intake openings to the total area of the bottom wall **2002** advantageously assists in minimizing any temperature variation across the flat top cooking surface of the griddle **1804** during cooking operations performed thereon.

[0114] As shown in FIGS. **18-21**, the grill **1800** further includes an example upper rim **1834** defined by or associated with the cookbox **1802**. In some examples, one or more portion(s) of the upper rim **1834** of the grill **1800** is/are formed by one or more portion(s) of the front wall **2004**, the rear wall **2006**, the right sidewall **2008**, and/or the left sidewall **2010** of the cookbox **1802**. In some examples, one or more portion(s) of the upper rim **1834** of the grill **1800** is/are additionally or alternatively formed by one or more portion(s) of the control panel **1808** and/or the frame **1826** of the grill **1800**. For example, as shown in FIGS. **18-21**, an example first portion **1836** of the upper rim **1834** (e.g., including a rear portion, a right side portion, and a left side portion of the upper rim **1834**) is defined by the cookbox **1802** of the grill **1800**, and an example second portion **1838** of the upper rim **1834** is defined by the control panel **1808** of the grill **1800**. The cookbox **1802**, the control panel **1808**, and/or the frame **1826** of the grill **1800** include a plurality of example griddle docking openings **2122** that are configured to receive and/or to otherwise engage corresponding ones of a plurality of griddle support members of the griddle **1804**. As further described below, the griddle docking openings **2122** and the griddle support members are configured to support the griddle **1804** of the grill **1800** on or above the cookbox **1802** of the grill **1800**, and/or on or above the upper rim **1834** of the grill **1800**.

[0115] FIG. **22** is a first perspective view showing the griddle **1804** of the grill **1800** of FIGS. **18** and **19** in isolation. FIG. **23** is a second perspective view of the griddle **1804** of FIG. **22**. As shown in FIGS. **22** and **23**, the griddle **1804** includes an example base **2202**. The base **2202** of the griddle **1804** includes an example flat top cooking surface **2204** and an example underside **2302** located opposite the flat top cooking surface **2204**. In the illustrated example of FIGS. **22** and **23**, the flat top cooking surface **2204** is a continuous, substantially flat and/or substantially planar surface that is free of openings aside from the upper waste disposal opening **1824** described herein. The flat top cooking surface **2204** is configured to support a variety of liquid, semi-liquid, and/or solid food items during a variety of cooking processes that may be performed on the griddle **1804**. In the illustrated example of FIGS. **22** and **23**, the flat top cooking surface **2204** and/or, more generally, the base **2202** of the griddle **1804** has a generally rectangular profile. In other examples, the flat top cooking surface **2204** and/or, more generally, the base **2202** of the griddle **1804** can instead have a non-rectangular profile (e.g., a circular profile, an oval profile, a triangular profile, a trapezoidal profile, etc.).

[0116] The griddle **1804** further includes an example front lip **2206**, and an example rear lip **2208**, an example right side lip **2210**, and an example left side lip **2212**. The rear lip **2208** of the griddle **1804** is located opposite the front lip **2206** of the griddle **1804**. The right side lip **2210** of the griddle **1804** extends between the front lip **2206** and the rear lip **2208** of the griddle **1804**. The left side lip **2212** of the griddle **1804** is located opposite the right side lip **2210** of the griddle **1804** and extends between the front lip **2206** and the rear lip **2208** of the griddle **1804**. The front lip **2206**, the

rear lip **2208**, the right side lip **2210**, and the left side lip **2212** of the griddle **1804** extend upwardly from the base **2202** of the griddle **1804** to provide vertical boundaries configured to prevent food items from sliding off the flat top cooking surface **2204** of the griddle **1804** during one or more cooking operation(s) performed thereon.

[0117] As shown in FIGS. **22** and **23**, the upper waste disposal opening **1824** formed in the base **2202** of the griddle **1804** extends completely through the base **2202** (e.g., from the flat top cooking surface **2204** of the base **2202** through to the underside **2302** of the base **2202**). The upper waste disposal opening **1824** facilitates the removal and/or disposal of liquid and/or solid cooking waste (e.g., grease, residue, particles, and/or other byproducts associated with a cooking operation) located on the flat top cooking surface **2204** of the griddle **1804**, as further described above. In the illustrated example of FIGS. **22** and **23**, the upper waste disposal opening **1824** is located proximate the front left corner of the base **2202** of the griddle **1804** (e.g., near the junction of the front lip **2206** and the left side lip **2212** of the griddle **1804**). In other examples, the upper waste disposal opening **1824** can instead be located proximate some other portion (e.g., the rear left corner, the front right corner, the rear right corner, etc.) of the base **2202** of the griddle **1804**.

[0118] The griddle **1804** further includes a plurality of example griddle support members **2304** (e.g., vertically oriented posts) coupled to and extending downwardly from the underside **2302** of the base **2202** of the griddle **1804**. The griddle support members **2304** are configured to support the underside **2302** and/or, more generally, the base **2202** of the griddle **1804** on or above the cookbox **1802** of the grill **1800**, and/or on or above the upper rim **1834** of the grill **1800**. In this regard, respective ones of the griddle support members **2304** of the griddle **1804** are configured to be received by and/or to otherwise engage corresponding ones of the griddle docking openings **2122** of the cookbox **1802** of the grill **1800**. In the illustrated example of FIGS. **22** and **23**, the griddle **1804** includes a total of four griddle support members **2304** configured to be received by and/or to otherwise engage a total of four griddle docking openings **2122**. In other examples, the griddle **1804** can instead include a different number (e.g., two, three, five, six, etc.) of griddle support members **2304**, and the cookbox **1802** can similarly include a corresponding different number (e.g., two, three, five, six, etc.) of griddle docking openings **2122**.

[0119] The griddle **1804** further includes a plurality of example anti-warping braces **2306** coupled to and extending downwardly from the underside **2302** of the base **2202** of the griddle **1804**. The anti-warping braces **2306** are configured to support, strengthen, and/or brace the underside **2302** and/or, more generally, the base **2202** of the griddle **1804** to limit and/or prevent any buckling and/or warping thereof. As shown in FIGS. **22** and **23**, the anti-warping braces **2306** are arranged in a right-to-left orientation along the underside **2302** of the base **2202** of the griddle **1804**. In other examples, the anti-warping braces **2306** can instead be arranged in a different orientation (e.g., a front-to-rear orientation, a diagonal orientation, etc.) along the underside **2302** of the base **2202** of the griddle **1804**. In the illustrated example of FIGS. **22** and **23**, the griddle **1804** includes a total of three anti-warping braces **2306**. In other examples, the griddle **1804** can instead include a different number (e.g., one, two, four, etc.) of anti-warping braces **2306**.

[0120] In the illustrated example of FIGS. **22** and **23**, the flat top cooking surface **2204** and/or, more generally, the base **2202** of the griddle **1804** has a width (e.g., measured between the right side lip **2210** to the left side lip **2212**) of approximately 36.0 inches, a depth (e.g., measured between the front lip **2206** and the rear lip **2208**) of approximately 21.0 inches, and an associated total area (e.g., calculated as width multiplied by depth) of approximately 756.0 square inches. In other examples, the width, the depth, and/or the total area of the flat top cooking surface **2204** can differ from the above-described dimensions.

[0121] FIG. **24** is a perspective view showing the first burner tube **1902** of the grill **1800** of FIGS. **18** and **19** in isolation. FIG. **25** is a top view of the first burner tube **1902** of FIG. **24**. FIG. **26** is a bottom view of the first burner tube **1902** of FIGS. **24** and **25**. FIG. **27** is a right side view of the first burner tube **1902** of FIGS. **24-26**. FIG. **28** is a cross-sectional view of the first burner tube

1902 of FIGS. **24-27** taken along section E-E of FIG. **25**. The construction of the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908** of the grill **1800** of FIGS. **18** and **19** is substantially identical to that of the first burner tube **1902** as described herein in connection with FIGS. **24-28**.

[0122] As shown in FIGS. **24-28**, the first burner tube **1902** includes an example first end **2402** (e.g., a front end), an example second end **2404** (e.g., a rear end), an example outer wall **2406** extending between the first end **2402** and the second end **2404**, and an example central axis **2408** extending between the first end **2402** and the second end **2404**. The first end **2402** of the first burner tube **1902** is partially open, with the first end **2402** being configured to receive an outlet of a control valve (e.g., the outlet of the first control valve **1914**) located proximate the front wall **2004** of the cookbox **1802**. The second end **2404** of the first burner tube **1902** is closed, with the second end **2404** being configured to be coupled to a mounting flange located proximate the rear wall **2006** of the cookbox **1802**. The first burner tube **1902** is accordingly configured to be arranged in a front-to-rear orientation when disposed in the cookbox **1802**. In the illustrated example of FIGS. **24-28**, the first burner tube **1902** is structured as a linear burner tube having a circular cross-sectional profile. In other example, the first burner tube **1902** can instead be structured as a linear burner tube having a different cross-sectional profile (e.g., a rectangular cross-sectional profile, an oval-shaped cross-sectional profile, a triangular cross-sectional profile, a trapezoidal cross-sectional profile, etc.). In still other examples, the first burner tube **1902** can instead be structured as a non-linear burner tube having one or more bend(s) and/or curve(s) formed therein.

[0123] The first burner tube **1902** of FIGS. **24-28** includes a plurality of ports formed in and extending through the outer wall **2406** of the first burner tube **1902**. As shown in FIGS. **24-28**, the plurality of ports includes an example first row of ports **2410**, an example second row of ports **2412**, an example third row of ports **2414**, and an example fourth row of ports **2416**. The first row of ports **2410** is arranged parallel to the central axis **2408**. The second row of ports **2412** is also arranged parallel to the central axis **2408**, with the second row of ports **2412** being spaced apart from (e.g., angularly displaced from) the first row of ports **2410** about the perimeter (e.g., about the circumference) of the outer wall **2406** of the first burner tube **1902**. The third row of ports **2414** is arranged perpendicular to the central axis **2408**. The fourth row of ports **2416** is also arranged perpendicular to the central axis **2408**, with the fourth row of ports **2416** being spaced apart from (e.g., longitudinally displaced from) the third row of ports **2414** along central axis **2408** of the first burner tube **1902**. In the illustrated example of FIGS. **24-28**, the third row of ports **2414** is located proximate the first end **2402** (e.g., the front end) of the first burner tube **1902**, and the fourth row of ports **2416** is located proximate the second end **2404** (e.g., the rear end) of the first burner tube **1902**. The first burner tube **1902** can include additional ports and/or additional rows of ports relative to those described above. For example, as shown in FIGS. **24-28**, the first burner tube **1902** further includes an example fifth row of ports **2418** arranged perpendicular to the central axis **2408**, with the fifth row of ports **2418** being spaced apart from and proximate to the third row of ports **2414**.

[0124] As shown in FIGS. **24-28**, the outer wall **2406** of the first burner tube **1902** has a circular cross-sectional profile. The second row of ports **2412** is angularly displaced from the first row of ports **2410** about the circumference of the outer wall **2406** by an example angle **2802** measuring approximately 120.0 degrees. In other examples, the angle **2802** at which the second row of ports **2412** is angularly displaced from the first row of ports **2410** about the circumference of the outer wall **2406** can be between 90.0 and 150.0 degrees. Satisfaction of the above-described range of the angle **2802** at which the second row of ports **2412** is angularly displaced from the first row of ports **2410** about the circumference of the outer wall **2406** advantageously improves the operating efficiency of the cooking engine of the grill **180**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the angle **2802** at which the second row of ports **2412** is angularly displaced from the first

row of ports **2410** about the circumference of the outer wall **2406** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0125] The first burner tube **1902** of FIGS. **24-28** is further configured such that the third row of ports **2414** of the first burner tube **1902** is spaced apart from the first end **2402** (e.g., the front end) of the first burner tube **1902** by an example distance **2502** of approximately 4.8 inches. In other examples, the distance **2502** between the third row of ports **2414** and the first end **2402** of the first burner tube **1902** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **2502** between the third row of ports **2414** and the first end **2402** of the first burner tube **1902** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **2502** between the third row of ports **2414** and the first end **2402** of the first burner tube **1902** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **184** during cooking operations performed thereon.

[0126] The first burner tube **1902** of FIGS. **24-28** is further configured such that the fourth row of ports **2416** of the first burner tube **1902** is spaced apart from the second end **2404** (e.g., the rear end) of the first burner tube **1902** by an example distance **2504** of approximately 3.0 inches. In other examples, the distance **2504** between the fourth row of ports **2416** and the second end **2404** of the first burner tube **1902** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **2504** between the fourth row of ports **2416** and the second end **2404** of the first burner tube **1902** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **2504** between the fourth row of ports **2416** and the second end **2404** of the first burner tube **1902** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0127] In the illustrated example of FIGS. **24-28**, the first burner tube **1902** has a length of approximately 21.2 inches measured along the central axis **2408** from the first end **2402** (e.g., the front end) to the second end **2404** (e.g., the rear end) of the first burner tube **1902**. In other examples, the first burner tube **1902** can instead have a length that is substantially greater than or substantially less than 21.2 inches. In the illustrated example of FIGS. **24-28**, the outer wall **2406** of the first burner tube **1902** has a diameter of approximately 1.0 inches measured across the circular cross-sectional profile of the outer wall **2406**. In other examples, the first burner tube **1902** can instead have a diameter that is substantially greater than or substantially less than 1.0 inches. In the illustrated example of FIGS. **24-28**, each one of the ports from among the first row of ports **2410** and the second row of ports **2412** respectively formed in and extending through the outer wall **2406** of the first burner tube **1902** has a diameter of approximately 0.08 inches measured across the circular outlet opening of the port. In other examples, one or more of the ports from among the first row of ports **2410** and the second row of ports **2412** respectively formed in and extending through the outer wall **2406** of the first burner tube **1902** can instead have a diameter that is substantially greater than or substantially less than 0.08 inches. In the illustrated example of FIGS. **24-28**, each one of the ports from among the third row of ports **2414**, the fourth row of ports **2416**, and the fifth row of ports **2418** respectively formed in and extending through the outer wall **2406** of the first burner tube **1902** has a diameter of approximately 0.07 inches measured across the circular outlet opening of the port. In other examples, one or more of the ports from among the third row of ports **2414**, the fourth row of ports **2416**, and the fifth row of ports **2418** respectively formed in and extending through the outer wall **2406** of the first burner tube **1902** can instead have a diameter that is substantially greater than or substantially less than 0.07 inches.

[0128] FIG. **29** is a front view showing the grill **1800** of FIGS. **18** and **19**, with the frame **1826**, the

lid **1830**, and certain other structural features of the grill **1800** omitted for enhanced viewability. FIG. **30** is a right side view of the grill **1800** of FIG. **29**. FIG. **31** is a top view of the grill **1800** of FIGS. **29** and **30**. FIG. **32** is a cross-sectional view of the grill **1800** of FIGS. **29-31** taken along section F-F of FIG. **31**. FIG. **33** is a cross-sectional view of the grill **1800** of FIGS. **29-32** taken along section G-G of FIG. **31**. FIG. **34** is a cross-sectional view of the grill **1800** of FIGS. **29-33** taken along section H-H of FIG. **29**.

[0129] As shown in FIGS. **29-34**, the griddle **1804** of the grill **1800** is disposed on or above the cookbox **1802** of the grill **1800**. More specifically, the griddle **1804** is removably positioned and/or docked on top of the cookbox **1802** via engagement of the griddle support members **2304** of the griddle **1804** and the griddle docking openings **2122** associated with the cookbox **1802**. As further shown in FIGS. **29-34**, the underside **2302** of the griddle **1804** is located above and spaced apart from the upper rim **1834** of the cookbox **1802** by an example vertical gap **2902** that is configured to vent heated air generated within the cookbox **1802** (e.g., heated air generated via the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and/or the fourth burner tube **1908**). In addition to extending between the upper rim **1834** of the cookbox **1802** and the underside **2302** of the griddle **1804**, the vertical gap **2902** further extends completely or substantially around a perimeter of the upper rim **1834** of the cookbox **1802**, and completely or substantially around a perimeter of the underside **2302** of the griddle **1804**. Configuring the vertical gap **2902** in this manner advantageously facilitates even and/or uniform venting of heat from the cookbox **1802** (e.g., along each of the front wall **2004**, the rear wall **2006**, the right sidewall **2008**, and the left sidewall **2010** of the cookbox **1802**) relative to the griddle **1804** (e.g., along the front lip **2206**, the rear lip **2208**, the right side lip **2210**, and the left side lip **2212** of the griddle **1804**).

[0130] In the illustrated example of FIGS. **29-34**, the vertical gap **2902** has a height of approximately 0.95 inches. In other examples, the vertical gap **2902** can have a height between 0.25 and 1.50 inches. Satisfaction of the above-described range of the height of the vertical gap **2902** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the height of the vertical gap **2902** advantageously assists in minimizing any temperature variation across the flat top cooking surface of the griddle **1804** during cooking operations performed thereon.

[0131] As shown in FIGS. **29-34**, the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) of the grill **1800** are disposed within the cookbox **1802** of the grill **1800** at a location between the bottom wall **2002** of the cookbox **1802** and the underside **2302** of the griddle **1804**. More specifically, the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) of the grill **1800** are disposed within the cookbox **1802** of the grill **1800** at a location between the bottom wall **2002** of the cookbox **1802** and the upper rim **1834** of the cookbox **1802**, with respective ones of the burner tubes being laterally spaced apart from one another and arranged in a front-to-rear orientation. As shown in FIGS. **29-34**, the first burner tube **1902** is laterally positioned to the right side of the first air intake opening **2102** and the second air intake opening **2104** respectively formed in the bottom wall **2002** of the cookbox **1802**. The second burner tube **1904** is laterally positioned to the left side of the first air intake opening **2102** and the second air intake opening **2104** respectively formed in the bottom wall **2002** of the cookbox **1802**, and to the right side of the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** respectively formed in the bottom wall **2002** of the cookbox **1802**. The second burner tube **1904** is accordingly located between the first air intake opening **2102** and the second air intake opening **2104** on the one hand and the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** on the other hand, such that the second burner tube **1904** is laterally aligned with the first solid portion **2116** of the bottom wall **2002** of the cookbox **1802**. The third burner tube **1906** is laterally positioned to the left side of the

third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** respectively formed in the bottom wall **2002** of the cookbox **1802**, and to the right side of the sixth air intake opening **2112** and the seventh air intake opening **2114** respectively formed in the bottom wall **2002** of the cookbox **1802**. The third burner tube **1906** is accordingly located between the third air intake opening **2106**, the fourth air intake opening **2108**, and the fifth air intake opening **2110** on the one hand and the sixth air intake opening **2112** and the seventh air intake opening **2114** on the other hand, such that the third burner tube **1906** is laterally aligned with the second solid portion **2118** of the bottom wall **2002** of the cookbox **1802**. The fourth burner tube **1908** is laterally positioned to the left side of the sixth air intake opening **2112** and the seventh air intake opening **2114** formed in the bottom wall **2002** of the cookbox **1802**.

[0132] In the illustrated example of FIGS. **29-34**, each of the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) is vertically positioned above the bottom wall **2002** of the cookbox **1802** by an example distance **3202** of approximately 2.5 inches measured vertically from the bottom wall **2002** of the cookbox **1802** to the central axis **2408** of the respective burner tube. In other examples, the distance **3202** at which each of the burner tubes is vertically positioned above the bottom wall **2002** of the cookbox **1802** can be between 1.5 and 3.5 inches. Satisfaction of the above-described range of the distance **3202** at which each of the burner tubes is vertically positioned above the bottom wall **2002** of the cookbox **1802** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **3202** at which each of the burner tubes is vertically positioned above the bottom wall **2002** of the cookbox **1802** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0133] In the illustrated example of FIGS. **29-34**, each of the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) is vertically positioned below the upper rim **1834** of the cookbox **1802** by an example distance **3204** of approximately 3.1 inches measured vertically from the upper rim **1834** of the cookbox **1802** to the central axis **2408** of the respective burner tube. In other examples, the distance **3204** at which each of the burner tubes is vertically positioned below the upper rim **1834** of the cookbox **1802** can be between 2.0 and 4.0 inches. Satisfaction of the above-described range of the distance **3204** at which each of the burner tubes is vertically positioned below the upper rim **1834** of the cookbox **1802** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **3204** at which each of the burner tubes is vertically positioned below the upper rim **1834** of the cookbox **1802** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0134] In the illustrated example of FIGS. **29-34**, each of the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) is vertically positioned below the underside **2302** of the griddle **1804** by an example distance **3206** of approximately 4.1 inches measured vertically from the underside **2302** of the griddle **1804** to the central axis **2408** of the respective burner tube. In other examples, the distance **3206** at which each of the burner tubes is vertically positioned below the underside **2302** of the griddle **1804** can be between 3.0 and 5.0 inches. Satisfaction of the above-described range of the distance **3206** at which each of the burner tubes is vertically positioned below the underside **2302** of the griddle **1804** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **3206** at which each of the burner tubes is vertically positioned below the underside **2302** of the griddle **1804** advantageously

assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0135] In the illustrated example of FIGS. **29-34**, the third row of ports **2414** formed on the outer wall **2406** of each of the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) is positioned inwardly from the front wall **2004** of the cookbox **1802** by an example distance **3302** of approximately 4.1 inches measured depthwise (e.g., along the central axis **2408** of the respective burner tube) from the front wall **2004** of the cookbox **1802** to the third row of ports **2414** of the respective burner tube. In other examples, the distance **3302** at which the third row of ports **2414** of the respective burner tube is positioned inwardly from the front wall **2004** of the cookbox **1802** can be between 1.0 and 6.0 inches.

Satisfaction of the above-described range of the distance **3302** at which the third row of ports **2414** of the respective burner tube is positioned inwardly from the front wall **2004** of the cookbox **1802** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **3302** at which the third row of ports **2414** of the respective burner tube is positioned inwardly from the front wall **2004** of the cookbox **1802** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0136] In the illustrated example of FIGS. **29-34**, the fourth row of ports **2416** formed on the outer wall **2406** of each of the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) is positioned inwardly from the rear wall **2006** of the cookbox **1802** by an example distance **3304** of approximately 2.2 inches measured depthwise (e.g., along the central axis **2408** of the respective burner tube) from the rear wall **2006** of the cookbox **1802** to the fourth row of ports **2416** of the respective burner tube. In other examples, the distance **3304** at which the fourth row of ports **2416** of the respective burner tube is positioned inwardly from the rear wall **2006** of the cookbox **1802** can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance **3304** at which the fourth row of ports **2416** of the respective burner tube is positioned inwardly from the rear wall **2006** of the cookbox **1802** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **3304** at which the fourth row of ports **2416** of the respective burner tube is positioned inwardly from the rear wall **2006** of the cookbox **1802** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0137] In the illustrated example of FIGS. **29-34**, the third row of ports **2414** formed on the outer wall **2406** of each of the burner tubes (e.g., the first burner tube **1902**, the second burner tube **1904**, the third burner tube **1906**, and the fourth burner tube **1908**) is positioned inwardly from the front lip **2206** of the griddle **1804** by an example distance **3306** of approximately 4.6 inches measured depthwise (e.g., along the central axis **2408** of the respective burner tube) from the front lip **2206** of the griddle **1804** to the third row of ports **2414** of the respective burner tube. In other examples, the distance **3306** at which the third row of ports **2414** of the respective burner tube is positioned inwardly from the front lip **2206** of the griddle **1804** can be between 1.0 and 6.0 inches.

Satisfaction of the above-described range of the distance **3306** at which the third row of ports **2414** of the respective burner tube is positioned inwardly from the front lip **2206** of the griddle **1804** advantageously improves the operating efficiency of the cooking engine of the grill **1800**, and also improves the heat distribution properties associated with the griddle **1804** of the grill **1800**. In this regard, satisfaction of the above-described range of the distance **3306** at which the third row of ports **2414** of the respective burner tube is positioned inwardly from the front lip **2206** of the griddle **1804** advantageously assists in minimizing any temperature variation across the flat top cooking surface **2204** of the griddle **1804** during cooking operations performed thereon.

[0138] In the illustrated example of FIGS. 29-34, the fourth row of ports 2416 formed on the outer wall 2406 of each of the burner tubes (e.g., the first burner tube 1902, the second burner tube 1904, the third burner tube 1906, and the fourth burner tube 1908) is positioned inwardly from the rear lip 2208 of the griddle 1804 by an example distance 3308 of approximately 3.3 inches measured depthwise (e.g., along the central axis 2408 of the respective burner tube) from the rear lip 2208 of the griddle 1804 to the fourth row of ports 2416 of the respective burner tube. In other examples, the distance 3308 at which the fourth row of ports 2416 of the respective burner tube is positioned inwardly from the rear lip 2208 of the griddle 1804 can be between 1.0 and 6.0 inches. Satisfaction of the above-described range of the distance 3308 at which the fourth row of ports 2416 of the respective burner tube is positioned inwardly from the rear lip 2208 of the griddle 1804 advantageously improves the operating efficiency of the cooking engine of the grill 1800, and also improves the heat distribution properties associated with the griddle 1804 of the grill 1800. In this regard, satisfaction of the above-described range of the distance 3308 at which the fourth row of ports 2416 of the respective burner tube is positioned inwardly from the rear lip 2208 of the griddle 1804 of the cookbox 1802 advantageously assists in minimizing any temperature variation across the flat top cooking surface 2204 of the griddle 1804 during cooking operations performed thereon.

[0139] The grill 1800 of FIGS. 18-34 is configured for optimum efficiency and optimum heat distribution during cooking operations performed on the flat top cooking surface 2204 of the griddle 1804. More specifically, the grill 1800 is configured to minimize any temperature variation across the flat top cooking surface 2204 of the griddle 1804 during cooking operations performed thereon. As shown in FIG. 31, the flat top cooking surface 2204 of the griddle 1804 includes an example central portion 3102 that is bounded and/or circumscribed by an example peripheral portion 3104. In the illustrated example, the peripheral portion 3104 of the flat top cooking surface 2204 extends approximately three inches inwardly from each of the four surrounding lips of the griddle 1804 (e.g., the front lip 2206, the rear lip 2208, the right side lip 2210, and the left side lip 2212). The grill 1800 of FIGS. 18-34 is configured such that the average temperature within the central portion 3102 of the flat top cooking surface 2204 of the griddle 1804 is approximately 365 degrees Fahrenheit (365° F.) when the burner tubes (e.g., the first burner tube 1902, the second burner tube 1904, the third burner tube 1906, and the fourth burner tube 1908) of the grill 1800 are adjusted to a LOW output setting, approximately 400 degrees Fahrenheit (400° F.) when the burner tubes of the grill 1800 are adjusted to a MEDIUM output setting, and approximately 535 degrees Fahrenheit (535° F.) when the burner tubes of the grill 1800 are adjusted to a HI output setting. As a result of implementing the innovative heat distribution features described above, the grill 1800 of FIGS. 18-34 is advantageously configured such that the average temperature variation between the central portion 3102 and the peripheral portion 3104 of the flat top cooking surface 2204 of the griddle 1804 is approximately 60 degrees Fahrenheit (60° F.) or less when the burner tubes of the grill 1800 are adjusted to a LOW output setting, approximately 90 degrees Fahrenheit (90° F.) or less when the burner tubes of the grill 1800 are adjusted to a MEDIUM output setting, and approximately 140 degrees Fahrenheit (140° F.) or less when the burner tubes of the grill 1800 are adjusted to a HI output setting. The above-described temperature variations are substantially less than those found in many commercially available flat top gas grills, some of which produce flat top cooking surface temperature variations exceeding two hundred degrees Fahrenheit (200° F.) when the burner tubes of the flat top gas grill are adjusted to a common output setting.

[0140] The following paragraphs provide various examples in relation to the disclosed flat top gas grills including cooking engines configured for optimum heat distribution.

[0141] Example 1 includes a grill. In Example 1, the grill includes a cookbox, a griddle, and a plurality of burner tubes. In Example 1, the cookbox includes a bottom wall, a front wall, a rear wall, a right sidewall, and a left sidewall. In Example 1, the front wall, the rear wall, the right sidewall, and the left sidewall extend upwardly from the bottom wall. In Example 1, the cookbox further includes a plurality of air intake openings extending through the bottom wall. In Example 1,

the griddle is disposed on or above the cookbox. In Example 1, the griddle includes a flat top cooking surface and an underside located opposite the flat top cooking surface. In Example 1, the burner tubes are disposed in the cookbox and located between the bottom wall of the cookbox and the underside of the griddle.

[0142] Example 2 includes the grill of Example 1. In Example 2, the bottom wall has a total area defined by a width extending between the right sidewall and the left sidewall and a depth extending between the front wall and the rear wall. In Example 2, each one of the plurality of air intake openings has an associated area, and the respective associated areas of the plurality of air intake openings collectively define an unoccupied area of the bottom wall. In Example 2, the unoccupied area is between 10.0 and 30.0 percent of the total area.

[0143] Example 3 includes the grill of Example 1. In Example 3, the plurality of burner tubes includes a first burner tube, a second burner tube laterally spaced apart from the first burner tube, and a third burner tube laterally spaced apart from both the first burner tube and the second burner tube, and the plurality of air intake openings includes a first air intake opening and a second air intake opening laterally spaced apart from the first air intake opening. In Example 3, the first air intake opening is laterally positioned between the first burner tube and the second burner tube, and the second air intake opening is laterally positioned between the second burner tube and the third burner tube.

[0144] Example 4 includes the grill of Example 3. In Example 4, the bottom wall includes a solid portion extending between the front wall and the rear wall and laterally aligned with the second burner tube. In Example 4, the solid portion extends between the first air intake opening and the second air intake opening. In Example 4, the solid portion is free of any opening extending through the bottom wall.

[0145] Example 5 includes the grill of Example 4. In Example 5, at least one of the first air intake opening or the second air intake opening has a rectangular shape.

[0146] Example 6 includes the grill of Example 1. In Example 6, the plurality of burner tubes includes a first burner tube, a second burner tube laterally spaced apart from the first burner tube, a third burner tube laterally spaced apart from both the first burner tube and the second burner tube, and a fourth burner tube laterally spaced apart from each one of the first burner tube, the second burner tube, and the third burner tube, and the plurality of air intake openings includes a first air intake opening, a second air intake opening laterally spaced apart from the first air intake opening, and a third air intake opening laterally spaced apart from both the first air intake opening and the second air intake opening. In Example 6, the first air intake opening is laterally positioned between the first burner tube and the second burner tube, the second air intake opening is laterally positioned between the second burner tube and the third burner tube, and the third air intake opening is laterally positioned between the third burner tube and the fourth burner tube.

[0147] Example 7 includes the grill of Example 6. In Example 7, the bottom wall includes a first solid portion extending between the front wall and the rear wall and laterally aligned with the second burner tube. In Example 7, the first solid portion extends between the first air intake opening and the second air intake opening. In Example 7, the first solid portion is free of any opening extending through the bottom wall. In Example 7, the bottom wall further includes a second solid portion laterally spaced apart from the first solid portion, the second solid portion extending between the front wall and the rear wall and laterally aligned with the third burner tube. In Example 7, the second solid portion extends between the second air intake opening and the third air intake opening. In Example 7, the second solid portion is free of any opening extending through the bottom wall.

[0148] Example 8 includes the grill of Example 7. In Example 8, at least one of the first air intake opening, the second air intake opening, or the third air intake opening has a rectangular shape.

[0149] Example 9 includes the grill of Example 7. In Example 9, at least one of the first air intake opening, the second air intake opening, or the third air intake opening has a trapezoidal shape.

[0150] Example 10 includes the grill of Example 1. In Example 10, the grill further comprises a vertical gap extending between an upper rim of the cookbox and the underside of the griddle. In Example 10, the vertical gap further extends around a perimeter of the upper rim of the cookbox and around a perimeter of the underside of the griddle. In Example 10, the vertical gap is configured to vent heated air generated within the cookbox.

[0151] Example 11 includes the grill of Example 10. In Example 11, the vertical gap has a height between 0.25 and 1.50 inches.

[0152] Example 12 includes the grill of Example 1. In Example 12, the plurality of burner tubes includes a linear burner tube having a central axis. In Example 12, the central axis is vertically positioned between 1.5 and 3.5 inches above the bottom wall of the cookbox.

[0153] Example 13 includes the grill of Example 12. In Example 13, the central axis is vertically positioned between 2.0 and 4.0 inches below an upper rim of the cookbox.

[0154] Example 14 includes the grill of Example 12. In Example 14, the central axis is vertically positioned between 3.0 and 5.0 inches below the underside of the griddle.

[0155] Example 15 includes the grill of Example 1. In Example 15, each one of the plurality of burner tubes is a linear burner tube having a central axis. In Example 15, the central axis of each one of the plurality of burner tubes is vertically positioned between 1.5 and 3.5 inches above the bottom wall of the cookbox, vertically positioned between 2.0 and 4.0 inches below an upper rim of the cookbox, and vertically positioned between 3.0 and 5.0 inches below the underside of the griddle.

[0156] Example 16 includes the grill of Example 1. In Example 16, the plurality of burner tubes includes a linear burner tube having a central axis, an outer wall, a first row of ports, and a second row of ports. In Example 16, the ports of the first row extend through the outer wall, and the first row of ports is arranged parallel to the central axis. In Example 16, the ports of the second row extend through the outer wall, and the second row of ports is arranged parallel to the central axis. In Example 16, the second row of ports is spaced apart from the first row of ports.

[0157] Example 17 includes the grill of Example 16. In Example 17, the outer wall of the linear burner tube has a circular cross-sectional profile. In Example 17, the second row of ports is angularly displaced from the first row of ports by an angle between 90.0 and 150.0 degrees.

[0158] Example 18 includes the grill of Example 16. In Example 18, the linear burner tube further includes a third row of ports and a fourth row of ports. In Example 18, the ports of the third row extend through the outer wall, and the third row of ports is arranged perpendicular to the central axis. In Example 18, the ports of the fourth row extend through the outer wall, and the third row of ports is arranged perpendicular to the central axis. In Example 18, the fourth row of ports is spaced apart from the third row of ports.

[0159] Example 19 includes the grill of Example 18. In Example 19, the third row of ports is located between 1.0 and 6.0 inches inwardly from the front wall of the cookbox.

[0160] Example 20 includes the grill of Example 19. In Example 20, the fourth row of ports is located between 1.0 and 6.0 inches inwardly from the rear wall of the cookbox.

[0161] Although certain example apparatus, systems, methods, and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all apparatus, systems, methods, and articles of manufacture fairly falling within the scope of the claims of this patent.

[0162] The following claims are hereby incorporated into this Detailed Description by this reference, with each claim standing on its own as a separate embodiment of the present disclosure.

Claims

1. A grill, comprising: a cookbox including a bottom wall and a lower waste disposal opening formed in and extending through the bottom wall; a griddle disposed on or above the cookbox, the

griddle including an upper waste disposal opening formed in and extending through a base of the griddle, the base including a flat top cooking surface and an underside located opposite the flat top cooking surface; a plurality of burner tubes disposed in the cookbox and located between the bottom wall of the cookbox and the underside of the griddle; a waste bin slidably coupled to the cookbox, the waste bin located below the lower waste disposal opening of the cookbox; and a waste disposal chute disposed within the cookbox, the waste disposal chute configured to receive waste from the upper waste disposal opening of the griddle and direct the received waste to pass through the lower waste disposal opening of the cookbox and into the waste bin.

2. The grill of claim 1, wherein the cookbox further includes a front wall, a rear wall, a right sidewall, a left sidewall, and a plurality of air intake openings, the front wall, the rear wall, the right sidewall, and the left sidewall extending upwardly from the bottom wall, the plurality of air intake openings formed in and extending through the bottom wall and configured to draw air from a surrounding atmosphere into the cookbox, wherein the plurality of burner tubes includes a first linear burner tube, a second linear burner tube laterally spaced apart from the first linear burner tube, and a third linear burner tube laterally spaced apart from both the first linear burner tube and the second linear burner tube, wherein the plurality of air intake openings includes a first air intake opening and a second air intake opening laterally spaced apart from the first air intake opening, wherein the first air intake opening is laterally positioned between the first linear burner tube and the second linear burner tube and the second air intake opening is laterally positioned between the second linear burner tube and the third linear burner tube, and wherein the bottom wall further includes a solid portion extending between the front wall and the rear wall and laterally aligned with the second linear burner tube, the solid portion extending between the first air intake opening and the second air intake opening, the solid portion being free of any opening extending through the bottom wall.

3. The grill of claim 2, wherein the bottom wall has a total area defined by a width extending between the right sidewall and the left sidewall and a depth extending between the front wall and the rear wall, wherein each one of the plurality of air intake openings has an associated area, wherein the respective associated areas of the plurality of air intake openings collectively define an unoccupied area of the bottom wall, and wherein the unoccupied area is between 10.0 and 30.0 percent of the total area.

4. The grill of claim 2, wherein at least one of the first air intake opening or the second air intake opening has a rectangular shape.

5. The grill of claim 2, wherein the plurality of burner tubes further includes a fourth linear burner tube laterally spaced apart from each one of the first linear burner tube, the second linear burner tube, and the third linear burner tube, wherein the plurality of air intake openings further includes a third air intake opening laterally spaced apart from both the first air intake opening and the second air intake opening, wherein the third air intake opening is laterally positioned between the third linear burner tube and the fourth linear burner tube.

6. The grill of claim 5, wherein the solid portion is a first solid portion, wherein the bottom wall further includes a second solid portion laterally spaced apart from the first solid portion, the second solid portion extending between the front wall and the rear wall and laterally aligned with the third linear burner tube, the second solid portion extending between the second air intake opening and the third air intake opening, the second solid portion being free of any opening extending through the bottom wall.

7. The grill of claim 6, wherein at least one of the first air intake opening, the second air intake opening, or the third air intake opening has a rectangular shape.

8. The grill of claim 6, wherein at least one of the first air intake opening, the second air intake opening, or the third air intake opening has a trapezoidal shape.

9. The grill of claim 2, further comprising a vertical gap extending between an upper rim of the cookbox and the underside of the griddle, the vertical gap further extending around a perimeter of

the upper rim of the cookbox and around a perimeter of the underside of the griddle, the vertical gap configured to vent heated air generated within the cookbox.

10. The grill of claim 9, wherein the vertical gap has a height between 0.25 and 1.50 inches.

11. The grill of claim 2, wherein each one of the plurality of burner tubes is a linear burner tube having a central axis, wherein the central axis of each one of the plurality of burner tubes is vertically positioned between 1.5 and 3.5 inches above the bottom wall of the cookbox.

12. The grill of claim 11, wherein the central axis of each one of the plurality of burner tubes is vertically positioned between 2.0 and 4.0 inches below an upper rim of the cookbox.

13. The grill of claim 11, wherein the central axis of each one of the plurality of burner tubes is vertically positioned between 3.0 and 5.0 inches below the underside of the griddle.

14. The grill of claim 2, wherein each one of the plurality of burner tubes is a linear burner tube having a central axis, wherein the central axis of each one of the plurality of burner tubes is vertically positioned between 1.5 and 3.5 inches above the bottom wall of the cookbox, vertically positioned between 2.0 and 4.0 inches below an upper rim of the cookbox, and vertically positioned between 3.0 and 5.0 inches below the underside of the griddle.

15. The grill of claim 2, wherein each one of the plurality of burner tubes includes a central axis, an outer wall having a circular cross-sectional profile, a first row of ports extending through the outer wall and arranged parallel to the central axis, and a second row of ports extending through the outer wall and arranged parallel to the central axis, the second row of ports angularly displaced from the first row of ports by an angle between 90.0 and 150.0 degrees.

16. The grill of claim 15, wherein each one of the plurality of burner tubes further includes a third row of ports extending through the outer wall and arranged perpendicular to the central axis, and a fourth row of ports extending through the outer wall and arranged perpendicular to the central axis, the fourth row of ports spaced apart from the third row of ports, the third row of ports and the fourth row of ports extending between the first row of ports and the second row of ports.

17. The grill of claim 16, wherein the third row of ports is located between 1.0 and 6.0 inches inwardly from the front wall of the cookbox.

18. The grill of claim 17, wherein the fourth row of ports is located between 1.0 and 6.0 inches inwardly from the rear wall of the cookbox.

19. The grill of claim 2, wherein a temperature variation across the flat top cooking surface of the griddle is one hundred fifty degrees Fahrenheit (150° F.) or less when each one of the plurality of burner tubes is adjusted to a common output setting.

20. The grill of claim 2, wherein a temperature variation across the flat top cooking surface of the griddle is sixty degrees Fahrenheit (60° F.) or less when each one of the plurality of burner tubes is adjusted to a low output setting, ninety degrees Fahrenheit (90° F.) or less when each one of the plurality of burner tubes is adjusted to a medium output setting, and one hundred forty degrees Fahrenheit (140° F.) or less when each one of the plurality of burner tubes is adjusted to a high output setting.
